

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.1	3	0.3	0	0	85.6
U2 - Beta(0.5, 0.5)	11.5	2.7	0.4	0	0	85.4
U2 - Beta(0.7, 0.3)	10.6	3.3	0.3	0.1	0	85.7
U2 - Beta(1, 1)	10.9	3	0.3	0.1	0	85.7
U3 - Beta(0.3, 0.7)	11.3	2.9	0.3	0	0	85.5
U3 - Beta(0.5, 0.5)	12.3	1.9	0.4	0	0	85.4
U3 - Beta(0.7, 0.3)	11.7	2.4	0.2	0	0	85.7
U3 - Beta(1, 1)	11.8	2.5	0.2	0	0	85.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.4	1	0.2	0	-
U2 - Beta(0.5, 0.5)	11.7	2.4	0.9	0.2	0	-
U2 - Beta(0.7, 0.3)	11.7	2.4	0.9	0.2	0	-
U2 - Beta(1, 1)	11.7	2.4	0.9	0.2	0	-
U3 - Beta(0.3, 0.7)	11.7	2.4	1	0.2	0	-
U3 - Beta(0.5, 0.5)	11.8	2.4	0.9	0.2	0	-
U3 - Beta(0.7, 0.3)	11.8	2.4	0.9	0.2	0	-
U3 - Beta(1, 1)	11.8	2.4	0.9	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	1.3	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	3.5	1.4	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	3.5	1.4	0.4	0.1	0	-
U2 - Beta(1, 1)	3.5	1.4	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	3.5	1.4	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	3.5	1.4	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	3.5	1.4	0.3	0.1	0	-
U3 - Beta(1, 1)	3.5	1.4	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						542.9
U2 - Beta(0.5, 0.5)						542.1
U2 - Beta(0.7, 0.3)						542.5
U2 - Beta(1, 1)						543.3
U3 - Beta(0.3, 0.7)						543.3
U3 - Beta(0.5, 0.5)						543
U3 - Beta(0.7, 0.3)						542.9
U3 - Beta(1, 1)						543.3
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	42.8	21.6	26.5	5.9	0.6	2.6
U2 - Beta(0.5, 0.5)	42.3	22.5	26.8	5.3	0.5	2.6
U2 - Beta(0.7, 0.3)	42.5	22.8	26.5	5.2	0.4	2.6
U2 - Beta(1, 1)	42.7	23.4	25.4	5.4	0.5	2.6
U3 - Beta(0.3, 0.7)	45	22.2	25	4.8	0.4	2.6
U3 - Beta(0.5, 0.5)	44.9	23.9	24	4.1	0.4	2.7
U3 - Beta(0.7, 0.3)	45.1	24	23.9	4	0.4	2.6
U3 - Beta(1, 1)	46	23.1	23.7	4.2	0.4	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	13.6	6	6.3	2.8	0.9	-
U2 - Beta(0.5, 0.5)	13.5	6.1	6.4	2.7	0.8	-
U2 - Beta(0.7, 0.3)	13.6	6.1	6.4	2.6	0.8	-
U2 - Beta(1, 1)	13.6	6.1	6.4	2.6	0.9	-
U3 - Beta(0.3, 0.7)	13.5	6	6.4	2.7	0.8	-
U3 - Beta(0.5, 0.5)	13.6	6	6.3	2.7	0.9	-
U3 - Beta(0.7, 0.3)	13.6	6.1	6.4	2.6	0.8	-
U3 - Beta(1, 1)	13.6	6.1	6.2	2.7	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	2.8	1.8	0.8	0.2	-
U2 - Beta(0.5, 0.5)	4.1	2.8	1.8	0.7	0.2	-
U2 - Beta(0.7, 0.3)	4.1	2.8	1.8	0.7	0.2	-
U2 - Beta(1, 1)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(0.3, 0.7)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(0.5, 0.5)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(0.7, 0.3)	4.1	2.8	1.8	0.7	0.2	-
U3 - Beta(1, 1)	4.1	2.8	1.7	0.8	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1093
U2 - Beta(0.5, 0.5)						1093.3
U2 - Beta(0.7, 0.3)						1092.4
U2 - Beta(1, 1)						1093.1
U3 - Beta(0.3, 0.7)						1093
U3 - Beta(0.5, 0.5)						1092.6
U3 - Beta(0.7, 0.3)						1092.3
U3 - Beta(1, 1)						1093.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19	41.2	16.5	4.6	0.5	18.2
U2 - Beta(0.5, 0.5)	18.6	41.7	16.3	4.8	0.5	18.1
U2 - Beta(0.7, 0.3)	19.3	42.9	15.1	4.1	0.4	18.2
U2 - Beta(1, 1)	18.9	42.8	15.4	4.2	0.5	18.2
U3 - Beta(0.3, 0.7)	19.4	43.8	14.3	4	0.3	18.2
U3 - Beta(0.5, 0.5)	19.9	44.4	13.7	3.5	0.4	18.1
U3 - Beta(0.7, 0.3)	19.5	45	13.9	3	0.3	18.3
U3 - Beta(1, 1)	20.4	45.2	12.7	3.1	0.3	18.3
Mean administrations						
U2 - Beta(0.3, 0.7)	10.4	8	5.4	2.4	0.8	-
U2 - Beta(0.5, 0.5)	10.4	8	5.3	2.5	0.8	-
U2 - Beta(0.7, 0.3)	10.4	8.1	5.3	2.4	0.8	-
U2 - Beta(1, 1)	10.4	8.1	5.3	2.4	0.8	-
U3 - Beta(0.3, 0.7)	10.4	8	5.4	2.4	0.8	-
U3 - Beta(0.5, 0.5)	10.5	8	5.3	2.4	0.8	-
U3 - Beta(0.7, 0.3)	10.4	8.1	5.3	2.4	0.8	-
U3 - Beta(1, 1)	10.4	8.1	5.3	2.4	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	3.4	1.5	0.8	0.2	-
U2 - Beta(0.5, 0.5)	3.1	3.4	1.5	0.9	0.2	-
U2 - Beta(0.7, 0.3)	3.1	3.5	1.5	0.8	0.2	-
U2 - Beta(1, 1)	3	3.5	1.5	0.8	0.2	-
U3 - Beta(0.3, 0.7)	3.1	3.4	1.5	0.8	0.2	-
U3 - Beta(0.5, 0.5)	3.1	3.4	1.5	0.8	0.3	-
U3 - Beta(0.7, 0.3)	3	3.5	1.5	0.8	0.2	-
U3 - Beta(1, 1)	3	3.5	1.5	0.8	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						987.6
U2 - Beta(0.5, 0.5)						986.9
U2 - Beta(0.7, 0.3)						986.8
U2 - Beta(1, 1)						987.4
U3 - Beta(0.3, 0.7)						987.3
U3 - Beta(0.5, 0.5)						987.3
U3 - Beta(0.7, 0.3)						987.7
U3 - Beta(1, 1)						986.8

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.2	11.9	27.7	39.9	7.8	4.5
U2 - Beta(0.5, 0.5)	8.2	11.6	27.8	40.4	7.3	4.7
U2 - Beta(0.7, 0.3)	9.4	11.5	26.1	41.5	7	4.5
U2 - Beta(1, 1)	10.5	11.2	26.1	40	7.8	4.4
U3 - Beta(0.3, 0.7)	11.4	13.7	25.6	37.4	7.5	4.4
U3 - Beta(0.5, 0.5)	12.3	12.8	24.6	39.3	6.6	4.4
U3 - Beta(0.7, 0.3)	13.8	11.8	24.4	39.8	5.7	4.5
U3 - Beta(1, 1)	15	11.6	23.4	39.1	6.3	4.6
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	4.9	7.5	7.8	4.5	-
U2 - Beta(0.5, 0.5)	4.6	4.9	7.5	7.8	4.5	-
U2 - Beta(0.7, 0.3)	4.7	4.9	7.4	8	4.3	-
U2 - Beta(1, 1)	4.7	4.9	7.3	7.8	4.6	-
U3 - Beta(0.3, 0.7)	4.6	4.8	7.5	7.9	4.5	-
U3 - Beta(0.5, 0.5)	4.7	4.8	7.5	7.8	4.5	-
U3 - Beta(0.7, 0.3)	4.7	4.9	7.5	8	4.2	-
U3 - Beta(1, 1)	4.7	4.9	7.3	7.8	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3	2.4	2.6	1.1	-
U2 - Beta(0.5, 0.5)	0.7	3	2.4	2.6	1.1	-
U2 - Beta(0.7, 0.3)	0.8	3	2.3	2.6	1.1	-
U2 - Beta(1, 1)	0.8	3	2.3	2.6	1.1	-
U3 - Beta(0.3, 0.7)	0.8	3	2.4	2.6	1.1	-
U3 - Beta(0.5, 0.5)	0.8	3	2.4	2.6	1.1	-
U3 - Beta(0.7, 0.3)	0.8	3	2.4	2.6	1.1	-
U3 - Beta(1, 1)	0.8	3	2.3	2.6	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1066.7
U2 - Beta(0.5, 0.5)						1065.4
U2 - Beta(0.7, 0.3)						1064.1
U2 - Beta(1, 1)						1066.1
U3 - Beta(0.3, 0.7)						1066
U3 - Beta(0.5, 0.5)						1064.8
U3 - Beta(0.7, 0.3)						1064.2
U3 - Beta(1, 1)						1065.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.2	32.9	27.4	21.2	7.2	8.1
U2 - Beta(0.5, 0.5)	3.8	36.1	25.6	19.6	6.8	8.1
U2 - Beta(0.7, 0.3)	4.8	35.5	26.5	20.2	4.9	8.1
U2 - Beta(1, 1)	4.3	36.9	26.7	18.3	5.7	8.1
U3 - Beta(0.3, 0.7)	5.1	38.3	25.7	18	5	7.9
U3 - Beta(0.5, 0.5)	5.3	40.2	24.9	17.6	3.9	8.1
U3 - Beta(0.7, 0.3)	6.3	43.5	23.7	14.8	3.6	8.1
U3 - Beta(1, 1)	7.3	42.4	23.6	15.3	3.3	8.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.8	6.6	7	5.9	4.2	-
U2 - Beta(0.5, 0.5)	4.8	6.8	6.8	5.8	4.2	-
U2 - Beta(0.7, 0.3)	4.8	7	6.9	5.8	4.1	-
U2 - Beta(1, 1)	4.8	6.9	6.8	5.8	4.1	-
U3 - Beta(0.3, 0.7)	4.8	6.6	7	5.8	4.2	-
U3 - Beta(0.5, 0.5)	4.8	6.7	6.9	5.8	4.3	-
U3 - Beta(0.7, 0.3)	4.8	7	6.8	5.7	4.2	-
U3 - Beta(1, 1)	4.8	7	6.8	5.7	4.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	3.6	2.2	2.2	1.5	-
U2 - Beta(0.5, 0.5)	0.8	3.6	2.1	2.2	1.5	-
U2 - Beta(0.7, 0.3)	0.8	3.7	2.1	2.2	1.5	-
U2 - Beta(1, 1)	0.8	3.7	2.1	2.2	1.5	-
U3 - Beta(0.3, 0.7)	0.8	3.6	2.2	2.2	1.5	-
U3 - Beta(0.5, 0.5)	0.8	3.6	2.2	2.2	1.5	-
U3 - Beta(0.7, 0.3)	0.8	3.7	2.1	2.2	1.5	-
U3 - Beta(1, 1)	0.8	3.7	2.1	2.2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1001.3
U2 - Beta(0.5, 0.5)						1000.2
U2 - Beta(0.7, 0.3)						1000.6
U2 - Beta(1, 1)						1000.3
U3 - Beta(0.3, 0.7)						1001.4
U3 - Beta(0.5, 0.5)						1000.4
U3 - Beta(0.7, 0.3)						999.8
U3 - Beta(1, 1)						1000

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	65.3	6.1	0.9	0.2	0	27.5
U2 - Beta(0.5, 0.5)	65.4	6.3	1	0.2	0	27.1
U2 - Beta(0.7, 0.3)	64.9	6.8	1	0.2	0	27.1
U2 - Beta(1, 1)	65.1	6.7	0.7	0.3	0	27.2
U3 - Beta(0.3, 0.7)	66.3	5.5	0.9	0	0	27.3
U3 - Beta(0.5, 0.5)	65.8	5.8	0.8	0.2	0	27.4
U3 - Beta(0.7, 0.3)	65.6	6.2	0.6	0.2	0	27.4
U3 - Beta(1, 1)	66.1	5.8	0.8	0.2	0	27.1
Mean administrations						
U2 - Beta(0.3, 0.7)	20.4	3.1	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	20.4	3.2	1.4	0.3	0	-
U2 - Beta(0.7, 0.3)	20.5	3.2	1.3	0.3	0	-
U2 - Beta(1, 1)	20.4	3.3	1.3	0.3	0	-
U3 - Beta(0.3, 0.7)	20.4	3.1	1.4	0.3	0.1	-
U3 - Beta(0.5, 0.5)	20.4	3.2	1.4	0.3	0	-
U3 - Beta(0.7, 0.3)	20.4	3.3	1.3	0.3	0	-
U3 - Beta(1, 1)	20.4	3.3	1.3	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.7	1.3	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	5.7	1.3	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	5.7	1.3	0.4	0.1	0	-
U2 - Beta(1, 1)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	5.7	1.3	0.4	0.1	0	-
U3 - Beta(1, 1)	5.7	1.3	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						977.4
U2 - Beta(0.5, 0.5)						977.2
U2 - Beta(0.7, 0.3)						979.1
U2 - Beta(1, 1)						979.3
U3 - Beta(0.3, 0.7)						976.5
U3 - Beta(0.5, 0.5)						978.5
U3 - Beta(0.7, 0.3)						978
U3 - Beta(1, 1)						978.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	34.9	39.5	15.4	5	1.7	3.5
U2 - Beta(0.5, 0.5)	34.7	40.6	14.4	5.5	1.2	3.6
U2 - Beta(0.7, 0.3)	34.2	40.8	14.5	6.2	0.8	3.5
U2 - Beta(1, 1)	34.1	40.5	15.3	5.3	1.4	3.4
U3 - Beta(0.3, 0.7)	35.7	40.7	14	4.7	1.3	3.6
U3 - Beta(0.5, 0.5)	34.7	44.5	11.6	4.7	0.8	3.7
U3 - Beta(0.7, 0.3)	34.9	43.1	13.2	4.7	0.5	3.6
U3 - Beta(1, 1)	35.4	43	12.7	4.4	0.8	3.7
Mean administrations						
U2 - Beta(0.3, 0.7)	13	7.4	5	2.6	1.3	-
U2 - Beta(0.5, 0.5)	12.9	7.5	5	2.6	1.2	-
U2 - Beta(0.7, 0.3)	13	7.5	5	2.7	1.2	-
U2 - Beta(1, 1)	13	7.5	5	2.6	1.3	-
U3 - Beta(0.3, 0.7)	12.9	7.3	5.1	2.7	1.3	-
U3 - Beta(0.5, 0.5)	13	7.5	5	2.6	1.2	-
U3 - Beta(0.7, 0.3)	13	7.6	5	2.6	1.2	-
U3 - Beta(1, 1)	13	7.5	5	2.7	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	3	1.3	0.8	0.4	-
U2 - Beta(0.5, 0.5)	3.8	3	1.2	0.8	0.4	-
U2 - Beta(0.7, 0.3)	3.8	3	1.2	0.8	0.4	-
U2 - Beta(1, 1)	3.8	3	1.2	0.8	0.4	-
U3 - Beta(0.3, 0.7)	3.8	3	1.3	0.8	0.4	-
U3 - Beta(0.5, 0.5)	3.8	3	1.2	0.8	0.4	-
U3 - Beta(0.7, 0.3)	3.8	3	1.2	0.8	0.3	-
U3 - Beta(1, 1)	3.8	3	1.2	0.8	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.6
U2 - Beta(0.5, 0.5)						1109.5
U2 - Beta(0.7, 0.3)						1109.6
U2 - Beta(1, 1)						1110.2
U3 - Beta(0.3, 0.7)						1109.7
U3 - Beta(0.5, 0.5)						1109.6
U3 - Beta(0.7, 0.3)						1109.6
U3 - Beta(1, 1)						1109.9
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.4	3	0.5	0	0	70.1
U2 - Beta(0.5, 0.5)	27.3	2.4	0.6	0	0	69.7
U2 - Beta(0.7, 0.3)	26.6	3	0.5	0	0	69.9
U2 - Beta(1, 1)	26.7	2.7	0.5	0	0	70.1
U3 - Beta(0.3, 0.7)	26.8	2.9	0.4	0	0	69.9
U3 - Beta(0.5, 0.5)	27.3	2.1	0.4	0	0	70.2
U3 - Beta(0.7, 0.3)	27.2	2.6	0.2	0	0	70
U3 - Beta(1, 1)	27.7	2.1	0.2	0	0	70
Mean administrations						
U2 - Beta(0.3, 0.7)	14.9	2.3	0.8	0.1	0	-
U2 - Beta(0.5, 0.5)	14.9	2.3	0.8	0.1	0	-
U2 - Beta(0.7, 0.3)	14.9	2.3	0.8	0.1	0	-
U2 - Beta(1, 1)	14.9	2.2	0.8	0.1	0	-
U3 - Beta(0.3, 0.7)	14.8	2.3	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	14.9	2.3	0.8	0.1	0	-
U3 - Beta(0.7, 0.3)	14.9	2.3	0.8	0.1	0	-
U3 - Beta(1, 1)	14.9	2.2	0.8	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	1.2	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	4.6	1.2	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	4.6	1.2	0.3	0.1	0	-
U2 - Beta(1, 1)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	4.5	1.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	4.6	1.2	0.3	0.1	0	-
U3 - Beta(1, 1)	4.6	1.2	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						657.1
U2 - Beta(0.5, 0.5)						657.1
U2 - Beta(0.7, 0.3)						654.6
U2 - Beta(1, 1)						654.8
U3 - Beta(0.3, 0.7)						656.8
U3 - Beta(0.5, 0.5)						656.3
U3 - Beta(0.7, 0.3)						655.4
U3 - Beta(1, 1)						655.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12	14.2	17.4	10	11.2	35.2
U2 - Beta(0.5, 0.5)	14.6	12	17.6	9.9	10.9	35
U2 - Beta(0.7, 0.3)	13.1	13.7	16.5	10.8	10.4	35.5
U2 - Beta(1, 1)	14.6	13.6	16.1	9.7	10.5	35.5
U3 - Beta(0.3, 0.7)	14.4	15.2	15.3	9.1	10.5	35.5
U3 - Beta(0.5, 0.5)	17.3	12.8	15.3	9.7	9.5	35.4
U3 - Beta(0.7, 0.3)	19.7	13.7	13.8	8.7	9	35.1
U3 - Beta(1, 1)	21.2	13.4	12.1	8.4	9.6	35.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	4.9	5.2	3.1	2.6	-
U2 - Beta(0.5, 0.5)	8.4	5	5.1	3.1	2.6	-
U2 - Beta(0.7, 0.3)	8.5	5.1	5	3.1	2.6	-
U2 - Beta(1, 1)	8.5	5.1	4.9	3.1	2.6	-
U3 - Beta(0.3, 0.7)	8.4	5	5.2	3.2	2.6	-
U3 - Beta(0.5, 0.5)	8.4	5	5.1	3.1	2.7	-
U3 - Beta(0.7, 0.3)	8.5	5.1	5	3.2	2.6	-
U3 - Beta(1, 1)	8.5	5.1	4.9	3	2.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	2.8	1.8	1.3	0.8	-
U2 - Beta(0.5, 0.5)	2.3	2.8	1.8	1.3	0.8	-
U2 - Beta(0.7, 0.3)	2.3	2.9	1.7	1.3	0.8	-
U2 - Beta(1, 1)	2.4	2.9	1.7	1.3	0.8	-
U3 - Beta(0.3, 0.7)	2.3	2.8	1.8	1.3	0.8	-
U3 - Beta(0.5, 0.5)	2.3	2.8	1.8	1.3	0.8	-
U3 - Beta(0.7, 0.3)	2.3	2.8	1.8	1.3	0.8	-
U3 - Beta(1, 1)	2.4	2.9	1.7	1.3	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						838.8
U2 - Beta(0.5, 0.5)						838.3
U2 - Beta(0.7, 0.3)						837
U2 - Beta(1, 1)						837.2
U3 - Beta(0.3, 0.7)						838.1
U3 - Beta(0.5, 0.5)						838.2
U3 - Beta(0.7, 0.3)						838
U3 - Beta(1, 1)						837.1

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	20.8	6.8	1.6	0.5	59.6
U2 - Beta(0.5, 0.5)	11	21.9	5.7	1.4	0.4	59.6
U2 - Beta(0.7, 0.3)	10.7	22.5	5.4	1.6	0.3	59.5
U2 - Beta(1, 1)	10.8	22.1	5.5	1.7	0.4	59.5
U3 - Beta(0.3, 0.7)	11.5	21.7	5.4	1.6	0.3	59.5
U3 - Beta(0.5, 0.5)	11.2	22.4	5.1	1.4	0.2	59.7
U3 - Beta(0.7, 0.3)	11.1	23.3	4.6	1.2	0.2	59.6
U3 - Beta(1, 1)	11.4	22.4	4.9	1.5	0.3	59.5
Mean administrations						
U2 - Beta(0.3, 0.7)	10.3	5.1	2.9	1	0.3	-
U2 - Beta(0.5, 0.5)	10.3	5.2	2.8	1	0.3	-
U2 - Beta(0.7, 0.3)	10.4	5.3	2.7	0.9	0.3	-
U2 - Beta(1, 1)	10.4	5.2	2.8	1	0.3	-
U3 - Beta(0.3, 0.7)	10.4	5.1	2.9	1	0.3	-
U3 - Beta(0.5, 0.5)	10.4	5.1	2.9	0.9	0.3	-
U3 - Beta(0.7, 0.3)	10.4	5.3	2.7	1	0.3	-
U3 - Beta(1, 1)	10.4	5.2	2.8	0.9	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	2.3	0.8	0.3	0.1	-
U2 - Beta(0.5, 0.5)	2.9	2.3	0.8	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3	2.4	0.8	0.3	0.1	-
U2 - Beta(1, 1)	3	2.3	0.8	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3	2.3	0.9	0.3	0.1	-
U3 - Beta(0.5, 0.5)	3	2.3	0.8	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3	2.4	0.8	0.3	0.1	-
U3 - Beta(1, 1)	3	2.4	0.8	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						714.7
U2 - Beta(0.5, 0.5)						714.8
U2 - Beta(0.7, 0.3)						714.3
U2 - Beta(1, 1)						714.5
U3 - Beta(0.3, 0.7)						713.9
U3 - Beta(0.5, 0.5)						714.6
U3 - Beta(0.7, 0.3)						713.8
U3 - Beta(1, 1)						715.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	73.7	12	0.6	0.1	0	13.6
U2 - Beta(0.5, 0.5)	71.4	13.2	1	0.2	0	14.2
U2 - Beta(0.7, 0.3)	72.2	13.4	0.7	0.2	0	13.5
U2 - Beta(1, 1)	71.7	13.8	0.6	0.1	0	13.8
U3 - Beta(0.3, 0.7)	74.6	10.9	0.5	0	0	14
U3 - Beta(0.5, 0.5)	73	12.7	0.4	0.1	0	13.8
U3 - Beta(0.7, 0.3)	72.8	12.5	0.6	0.1	0	14
U3 - Beta(1, 1)	72.9	12.8	0.6	0.1	0	13.6
Mean administrations						
U2 - Beta(0.3, 0.7)	20.2	5.2	2	0.3	0	-
U2 - Beta(0.5, 0.5)	20.2	5.2	2	0.3	0	-
U2 - Beta(0.7, 0.3)	20.3	5.2	1.9	0.3	0	-
U2 - Beta(1, 1)	20.3	5.2	1.9	0.3	0	-
U3 - Beta(0.3, 0.7)	20.2	5.2	2	0.3	0	-
U3 - Beta(0.5, 0.5)	20.2	5.2	2	0.3	0	-
U3 - Beta(0.7, 0.3)	20.3	5.2	1.9	0.3	0	-
U3 - Beta(1, 1)	20.4	5.2	1.9	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	2.4	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	6.4	2.4	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	6.4	2.4	0.7	0.1	0	-
U2 - Beta(1, 1)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	6.4	2.4	0.7	0.1	0	-
U3 - Beta(1, 1)	6.5	2.4	0.6	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						996
U2 - Beta(0.5, 0.5)						994.3
U2 - Beta(0.7, 0.3)						995.8
U2 - Beta(1, 1)						995.6
U3 - Beta(0.3, 0.7)						994.7
U3 - Beta(0.5, 0.5)						996
U3 - Beta(0.7, 0.3)						993.5
U3 - Beta(1, 1)						994.3
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14	18.2	16.2	35.4	9.6	6.6
U2 - Beta(0.5, 0.5)	16.1	19.2	14.7	34.1	9.3	6.6
U2 - Beta(0.7, 0.3)	16.3	19.4	14.5	34.3	8.9	6.6
U2 - Beta(1, 1)	17.2	19.7	12.6	34.8	9.2	6.5
U3 - Beta(0.3, 0.7)	20.5	18.8	13.1	31.6	9.2	6.8
U3 - Beta(0.5, 0.5)	22.4	18.2	10.4	33.1	9.1	6.8
U3 - Beta(0.7, 0.3)	23.7	19.8	9.1	32	8.6	6.8
U3 - Beta(1, 1)	24.9	20	8.4	31.4	8.6	6.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	5.5	6.2	7.3	4.7	-
U2 - Beta(0.5, 0.5)	5.2	5.6	6.2	7.3	4.7	-
U2 - Beta(0.7, 0.3)	5.3	5.6	6.1	7.2	4.7	-
U2 - Beta(1, 1)	5.3	5.6	6.1	7.2	4.7	-
U3 - Beta(0.3, 0.7)	5.2	5.4	6.3	7.3	4.8	-
U3 - Beta(0.5, 0.5)	5.2	5.5	6.2	7.2	4.7	-
U3 - Beta(0.7, 0.3)	5.3	5.7	6.1	7.2	4.6	-
U3 - Beta(1, 1)	5.4	5.7	6	7.2	4.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	3.3	2	2.7	1.1	-
U2 - Beta(0.5, 0.5)	1	3.3	2.1	2.7	1.1	-
U2 - Beta(0.7, 0.3)	1.1	3.3	2	2.7	1.1	-
U2 - Beta(1, 1)	1.1	3.3	2	2.7	1.1	-
U3 - Beta(0.3, 0.7)	1	3.2	2.1	2.7	1.1	-
U3 - Beta(0.5, 0.5)	1	3.3	2	2.7	1.1	-
U3 - Beta(0.7, 0.3)	1.1	3.4	2	2.7	1.1	-
U3 - Beta(1, 1)	1.1	3.4	2	2.7	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1031.9
U2 - Beta(0.5, 0.5)						1031.9
U2 - Beta(0.7, 0.3)						1029.9
U2 - Beta(1, 1)						1031
U3 - Beta(0.3, 0.7)						1032.1
U3 - Beta(0.5, 0.5)						1031.3
U3 - Beta(0.7, 0.3)						1029.5
U3 - Beta(1, 1)						1030

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.4	22.9	7.5	1.2	0.3	54.7
U2 - Beta(0.5, 0.5)	13.8	22.8	7.3	1.3	0.2	54.6
U2 - Beta(0.7, 0.3)	12.7	24.5	6.7	1.1	0.3	54.7
U2 - Beta(1, 1)	13.4	23.3	7	1.2	0.3	54.8
U3 - Beta(0.3, 0.7)	14.4	23.4	6.4	1	0.2	54.6
U3 - Beta(0.5, 0.5)	14.9	22.7	6.4	1	0.2	54.8
U3 - Beta(0.7, 0.3)	14.7	23.5	5.7	1.1	0.2	54.8
U3 - Beta(1, 1)	14.6	23.5	5.6	1.4	0.3	54.6
Mean administrations						
U2 - Beta(0.3, 0.7)	10.3	5.6	3.1	1.1	0.3	-
U2 - Beta(0.5, 0.5)	10.3	5.6	3.2	1.1	0.3	-
U2 - Beta(0.7, 0.3)	10.3	5.7	3.1	1	0.3	-
U2 - Beta(1, 1)	10.3	5.7	3	1.1	0.3	-
U3 - Beta(0.3, 0.7)	10.4	5.6	3.2	1.1	0.3	-
U3 - Beta(0.5, 0.5)	10.3	5.6	3.2	1.1	0.3	-
U3 - Beta(0.7, 0.3)	10.3	5.7	3.1	1.1	0.3	-
U3 - Beta(1, 1)	10.3	5.7	3	1.1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	2.7	1	0.4	0.1	-
U2 - Beta(0.5, 0.5)	2.9	2.7	1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	2.9	2.7	1	0.4	0.1	-
U2 - Beta(1, 1)	3	2.7	1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3	2.7	1	0.4	0.1	-
U3 - Beta(0.5, 0.5)	3	2.7	1	0.5	0.1	-
U3 - Beta(0.7, 0.3)	3	2.7	1	0.4	0.1	-
U3 - Beta(1, 1)	2.9	2.7	1	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						732.2
U2 - Beta(0.5, 0.5)						731.8
U2 - Beta(0.7, 0.3)						730.6
U2 - Beta(1, 1)						730.1
U3 - Beta(0.3, 0.7)						732.1
U3 - Beta(0.5, 0.5)						730.7
U3 - Beta(0.7, 0.3)						730.5
U3 - Beta(1, 1)						731

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.1	19.4	18.3	19.6	6.6	18
U2 - Beta(0.5, 0.5)	20.9	19.6	15.9	20.5	5.3	17.8
U2 - Beta(0.7, 0.3)	21.5	19.8	15.6	19.3	6	17.8
U2 - Beta(1, 1)	21.9	20.7	14.1	19.7	5.6	18
U3 - Beta(0.3, 0.7)	25	20	12.7	18.5	5.5	18.3
U3 - Beta(0.5, 0.5)	27.9	18.8	12.9	17.6	4.7	18.1
U3 - Beta(0.7, 0.3)	29.4	20	11.7	16.7	4.5	17.7
U3 - Beta(1, 1)	32.5	17.9	9.8	17	4.7	18.1
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	5.9	6.3	5.1	3	-
U2 - Beta(0.5, 0.5)	7	6	6.3	5	2.9	-
U2 - Beta(0.7, 0.3)	7	6.2	6.1	5	2.9	-
U2 - Beta(1, 1)	7.1	6.2	6	5	2.9	-
U3 - Beta(0.3, 0.7)	7	5.9	6.3	4.9	3.1	-
U3 - Beta(0.5, 0.5)	7.1	6	6.2	4.9	3	-
U3 - Beta(0.7, 0.3)	7.1	6.2	6	5	3	-
U3 - Beta(1, 1)	7.1	6.2	6	5	2.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	3.4	2.2	2	0.9	-
U2 - Beta(0.5, 0.5)	1.8	3.4	2.2	2	0.9	-
U2 - Beta(0.7, 0.3)	1.8	3.5	2.1	2	0.9	-
U2 - Beta(1, 1)	1.8	3.5	2.1	2	0.9	-
U3 - Beta(0.3, 0.7)	1.8	3.4	2.2	1.9	1	-
U3 - Beta(0.5, 0.5)	1.8	3.4	2.2	2	0.9	-
U3 - Beta(0.7, 0.3)	1.8	3.5	2.1	2	0.9	-
U3 - Beta(1, 1)	1.8	3.5	2.1	2	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						932.5
U2 - Beta(0.5, 0.5)						931.1
U2 - Beta(0.7, 0.3)						930
U2 - Beta(1, 1)						930.1
U3 - Beta(0.3, 0.7)						931.7
U3 - Beta(0.5, 0.5)						930.9
U3 - Beta(0.7, 0.3)						930.4
U3 - Beta(1, 1)						927.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	43.1	29.1	7.4	1.9	0	18.5
U2 - Beta(0.5, 0.5)	44.5	27.5	8.1	1.4	0	18.5
U2 - Beta(0.7, 0.3)	44.7	28.5	6.9	1.6	0	18.3
U2 - Beta(1, 1)	45.8	27.6	6.8	1.4	0.1	18.3
U3 - Beta(0.3, 0.7)	49.2	27.3	4.3	0.7	0	18.5
U3 - Beta(0.5, 0.5)	50	26.1	4.6	1	0	18.3
U3 - Beta(0.7, 0.3)	51.4	24.4	4.8	0.9	0	18.5
U3 - Beta(1, 1)	53.1	23.6	3.9	1	0	18.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.9	7.5	5.3	1.9	0.6	-
U2 - Beta(0.5, 0.5)	12	7.6	5.1	1.9	0.6	-
U2 - Beta(0.7, 0.3)	12.1	7.7	5	2	0.6	-
U2 - Beta(1, 1)	12.2	7.6	5	1.9	0.6	-
U3 - Beta(0.3, 0.7)	12	7.6	5.2	1.9	0.6	-
U3 - Beta(0.5, 0.5)	12	7.7	5	1.9	0.6	-
U3 - Beta(0.7, 0.3)	12.1	7.6	5	2	0.6	-
U3 - Beta(1, 1)	12.2	7.6	4.9	1.9	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	3.8	1.9	0.9	0.2	-
U2 - Beta(0.5, 0.5)	3.8	3.8	1.8	0.9	0.2	-
U2 - Beta(0.7, 0.3)	3.8	3.9	1.8	0.9	0.2	-
U2 - Beta(1, 1)	3.9	3.8	1.8	0.9	0.2	-
U3 - Beta(0.3, 0.7)	3.8	3.8	1.8	0.9	0.2	-
U3 - Beta(0.5, 0.5)	3.8	3.9	1.8	0.9	0.2	-
U3 - Beta(0.7, 0.3)	3.8	3.8	1.8	0.9	0.2	-
U3 - Beta(1, 1)	3.9	3.8	1.7	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						915.5
U2 - Beta(0.5, 0.5)						916.1
U2 - Beta(0.7, 0.3)						915.9
U2 - Beta(1, 1)						916
U3 - Beta(0.3, 0.7)						915
U3 - Beta(0.5, 0.5)						915.8
U3 - Beta(0.7, 0.3)						916
U3 - Beta(1, 1)						915.5

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	24.7	31	17.5	15.5	11.3	0
U2 - Beta(0.5, 0.5)	26.4	30.6	17.7	15.1	10.2	0
U2 - Beta(0.7, 0.3)	26	32.9	16.4	14.5	10.2	0
U2 - Beta(1, 1)	27.9	32.1	16.8	14.4	8.8	0
U3 - Beta(0.3, 0.7)	31.9	35.2	14.9	11.6	6.4	0
U3 - Beta(0.5, 0.5)	31.9	36.1	13.8	11.4	6.8	0
U3 - Beta(0.7, 0.3)	34.2	36.9	13.5	10.4	5	0
U3 - Beta(1, 1)	35	37.6	12.9	10.3	4.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	5.6	6	5.5	5.9	7	-
U2 - Beta(0.5, 0.5)	5.6	6.1	5.5	6	6.8	-
U2 - Beta(0.7, 0.3)	5.6	6.2	5.5	6	6.7	-
U2 - Beta(1, 1)	5.6	6.2	5.5	6	6.8	-
U3 - Beta(0.3, 0.7)	5.6	5.9	5.5	5.9	7.1	-
U3 - Beta(0.5, 0.5)	5.6	6.1	5.5	5.9	6.9	-
U3 - Beta(0.7, 0.3)	5.6	6.2	5.5	5.9	6.8	-
U3 - Beta(1, 1)	5.6	6.2	5.5	5.9	6.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.3	1.9	2.6	2.7	-
U2 - Beta(0.5, 0.5)	1.1	3.3	1.9	2.7	2.6	-
U2 - Beta(0.7, 0.3)	1.1	3.4	1.9	2.7	2.6	-
U2 - Beta(1, 1)	1.1	3.4	1.9	2.7	2.6	-
U3 - Beta(0.3, 0.7)	1.1	3.3	1.9	2.6	2.7	-
U3 - Beta(0.5, 0.5)	1.1	3.4	1.9	2.7	2.7	-
U3 - Beta(0.7, 0.3)	1.1	3.4	1.9	2.6	2.7	-
U3 - Beta(1, 1)	1.1	3.4	1.9	2.7	2.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.8
U2 - Beta(0.5, 0.5)						1006.6
U2 - Beta(0.7, 0.3)						1007
U2 - Beta(1, 1)						1007.3
U3 - Beta(0.3, 0.7)						1007.2
U3 - Beta(0.5, 0.5)						1006.6
U3 - Beta(0.7, 0.3)						1006.6
U3 - Beta(1, 1)						1006.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	35.6	54.7	9.1	0.2	0	0.4
U2 - Beta(0.5, 0.5)	35.1	55.6	8.8	0.1	0	0.4
U2 - Beta(0.7, 0.3)	34.4	55.4	9.6	0.1	0	0.5
U2 - Beta(1, 1)	34.5	56.1	8.8	0.2	0	0.4
U3 - Beta(0.3, 0.7)	37	54.8	7.8	0	0	0.4
U3 - Beta(0.5, 0.5)	37.2	54.7	7.6	0.1	0	0.4
U3 - Beta(0.7, 0.3)	35.8	55.4	8.1	0.3	0	0.4
U3 - Beta(1, 1)	37	54.8	7.7	0.1	0	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	10.7	6.1	1.3	0.2	-
U2 - Beta(0.5, 0.5)	11.7	10.8	6	1.2	0.2	-
U2 - Beta(0.7, 0.3)	11.8	10.8	5.9	1.2	0.1	-
U2 - Beta(1, 1)	11.6	10.8	6.1	1.2	0.1	-
U3 - Beta(0.3, 0.7)	11.7	10.6	6.2	1.2	0.2	-
U3 - Beta(0.5, 0.5)	11.7	10.7	6	1.3	0.1	-
U3 - Beta(0.7, 0.3)	11.8	10.8	5.9	1.2	0.1	-
U3 - Beta(1, 1)	11.7	10.7	6	1.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	3.7	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	3.1	3.7	1.4	0.4	0	-
U2 - Beta(0.7, 0.3)	3.1	3.7	1.4	0.4	0	-
U2 - Beta(1, 1)	3	3.7	1.4	0.4	0	-
U3 - Beta(0.3, 0.7)	3.1	3.7	1.4	0.4	0	-
U3 - Beta(0.5, 0.5)	3.1	3.7	1.4	0.4	0	-
U3 - Beta(0.7, 0.3)	3.1	3.7	1.3	0.4	0	-
U3 - Beta(1, 1)	3.1	3.7	1.4	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1178.4
U2 - Beta(0.5, 0.5)						1178
U2 - Beta(0.7, 0.3)						1176.4
U2 - Beta(1, 1)						1178.2
U3 - Beta(0.3, 0.7)						1177.8
U3 - Beta(0.5, 0.5)						1178.3
U3 - Beta(0.7, 0.3)						1177
U3 - Beta(1, 1)						1178.1

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.9	14.4	13.3	21.2	41.3	0.9
U2 - Beta(0.5, 0.5)	9.9	13.3	12.8	20.4	42.7	0.9
U2 - Beta(0.7, 0.3)	10.7	13.8	12.1	19.9	42.6	0.9
U2 - Beta(1, 1)	11.8	14.2	11.7	19.7	41.7	0.9
U3 - Beta(0.3, 0.7)	12.8	16.7	12.2	18.4	39	0.9
U3 - Beta(0.5, 0.5)	14.2	16.8	11.8	18.9	37.5	0.8
U3 - Beta(0.7, 0.3)	16	17.3	11.9	16.1	37.8	0.9
U3 - Beta(1, 1)	18.4	16.6	11.3	15.4	37.4	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	4	4.9	6.1	11.2	-
U2 - Beta(0.5, 0.5)	3.7	4.1	5	6	10.9	-
U2 - Beta(0.7, 0.3)	3.7	4.3	5	6	10.9	-
U2 - Beta(1, 1)	3.7	4.2	5	6.1	10.8	-
U3 - Beta(0.3, 0.7)	3.6	4	4.9	6	11.2	-
U3 - Beta(0.5, 0.5)	3.7	4.1	5	6.1	11	-
U3 - Beta(0.7, 0.3)	3.8	4.3	4.9	6.1	10.8	-
U3 - Beta(1, 1)	3.7	4.3	4.9	6	10.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.8	1.6	2.7	3.1	-
U2 - Beta(0.5, 0.5)	0.3	2.8	1.7	2.7	3.1	-
U2 - Beta(0.7, 0.3)	0.3	2.9	1.7	2.6	3.1	-
U2 - Beta(1, 1)	0.4	2.9	1.7	2.7	3	-
U3 - Beta(0.3, 0.7)	0.3	2.8	1.6	2.6	3.1	-
U3 - Beta(0.5, 0.5)	0.3	2.8	1.7	2.7	3.1	-
U3 - Beta(0.7, 0.3)	0.4	2.9	1.6	2.7	3.1	-
U3 - Beta(1, 1)	0.3	2.9	1.6	2.6	3.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1064.9
U2 - Beta(0.5, 0.5)						1062.3
U2 - Beta(0.7, 0.3)						1059.8
U2 - Beta(1, 1)						1060.8
U3 - Beta(0.3, 0.7)						1066
U3 - Beta(0.5, 0.5)						1062.2
U3 - Beta(0.7, 0.3)						1060.2
U3 - Beta(1, 1)						1060.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.7	49.4	6.8	0.2	0	17.9
U2 - Beta(0.5, 0.5)	26	48.4	6.9	0.3	0	18.4
U2 - Beta(0.7, 0.3)	26.1	48.9	6.5	0.3	0	18.2
U2 - Beta(1, 1)	26.4	47.9	7.1	0.4	0	18.2
U3 - Beta(0.3, 0.7)	26.9	48.4	6	0.2	0	18.5
U3 - Beta(0.5, 0.5)	27	48.2	6.3	0.3	0	18.2
U3 - Beta(0.7, 0.3)	28	48	5.6	0.3	0	18.1
U3 - Beta(1, 1)	28.9	46.9	5.5	0.4	0	18.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	10.5	5.5	1	0.1	-
U2 - Beta(0.5, 0.5)	10	10.6	5.3	1	0.1	-
U2 - Beta(0.7, 0.3)	10	10.7	5.2	1	0.1	-
U2 - Beta(1, 1)	10	10.5	5.3	1	0.1	-
U3 - Beta(0.3, 0.7)	9.9	10.5	5.5	1	0.1	-
U3 - Beta(0.5, 0.5)	10	10.6	5.3	1	0.1	-
U3 - Beta(0.7, 0.3)	9.9	10.7	5.2	1	0.1	-
U3 - Beta(1, 1)	10	10.6	5.3	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.6	1.6	0.4	0	-
U2 - Beta(0.5, 0.5)	2.9	4.6	1.6	0.4	0	-
U2 - Beta(0.7, 0.3)	2.9	4.6	1.5	0.4	0	-
U2 - Beta(1, 1)	2.9	4.6	1.6	0.4	0	-
U3 - Beta(0.3, 0.7)	2.8	4.6	1.6	0.4	0	-
U3 - Beta(0.5, 0.5)	2.9	4.6	1.6	0.4	0	-
U3 - Beta(0.7, 0.3)	2.9	4.7	1.5	0.4	0	-
U3 - Beta(1, 1)	2.9	4.6	1.5	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						965.3
U2 - Beta(0.5, 0.5)						963
U2 - Beta(0.7, 0.3)						964.2
U2 - Beta(1, 1)						963.6
U3 - Beta(0.3, 0.7)						965.4
U3 - Beta(0.5, 0.5)						962.6
U3 - Beta(0.7, 0.3)						964.5
U3 - Beta(1, 1)						962.8
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.3	50.8	26	3.6	0	9.3
U2 - Beta(0.5, 0.5)	11.1	51.8	24.2	3.7	0.2	9
U2 - Beta(0.7, 0.3)	11.2	49.8	25.6	4.1	0.1	9.2
U2 - Beta(1, 1)	11.6	50.6	24.6	4	0.1	9.1
U3 - Beta(0.3, 0.7)	14.2	50.5	22.8	3.3	0.1	9.1
U3 - Beta(0.5, 0.5)	14.5	50.8	22.4	3.3	0.2	8.8
U3 - Beta(0.7, 0.3)	14.5	51.4	20.7	4	0.1	9.3
U3 - Beta(1, 1)	16.9	51.5	18.5	3.6	0.1	9.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	9.8	8.5	3.3	1	-
U2 - Beta(0.5, 0.5)	5.7	9.9	8.5	3.2	1	-
U2 - Beta(0.7, 0.3)	5.7	10.1	8.4	3.2	1	-
U2 - Beta(1, 1)	5.8	10	8.2	3.3	1	-
U3 - Beta(0.3, 0.7)	5.7	9.6	8.7	3.3	1	-
U3 - Beta(0.5, 0.5)	5.8	9.8	8.5	3.2	1	-
U3 - Beta(0.7, 0.3)	5.8	10	8.3	3.3	1	-
U3 - Beta(1, 1)	5.7	10	8.3	3.3	1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	4.7	2.7	1.3	0.4	-
U2 - Beta(0.5, 0.5)	1.2	4.8	2.6	1.3	0.4	-
U2 - Beta(0.7, 0.3)	1.2	4.8	2.6	1.3	0.4	-
U2 - Beta(1, 1)	1.2	4.8	2.6	1.3	0.4	-
U3 - Beta(0.3, 0.7)	1.2	4.7	2.7	1.3	0.4	-
U3 - Beta(0.5, 0.5)	1.2	4.7	2.7	1.3	0.4	-
U3 - Beta(0.7, 0.3)	1.2	4.8	2.6	1.3	0.4	-
U3 - Beta(1, 1)	1.2	4.8	2.6	1.3	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						993.9
U2 - Beta(0.5, 0.5)						992.8
U2 - Beta(0.7, 0.3)						992.9
U2 - Beta(1, 1)						993.6
U3 - Beta(0.3, 0.7)						994.8
U3 - Beta(0.5, 0.5)						993.8
U3 - Beta(0.7, 0.3)						991.5
U3 - Beta(1, 1)						992.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.6	0.1	0	0	0	62.3
U2 - Beta(0.5, 0.5)	38.1	0.2	0	0	0	61.7
U2 - Beta(0.7, 0.3)	37.5	0.2	0	0	0	62.3
U2 - Beta(1, 1)	37.9	0.2	0	0	0	61.9
U3 - Beta(0.3, 0.7)	37.7	0.1	0	0	0	62.2
U3 - Beta(0.5, 0.5)	37.8	0.2	0	0	0	62
U3 - Beta(0.7, 0.3)	37.8	0.3	0	0	0	61.9
U3 - Beta(1, 1)	37.2	0.1	0	0	0	62.7
Mean administrations						
U2 - Beta(0.3, 0.7)	18.7	1.4	0.2	0	0	-
U2 - Beta(0.5, 0.5)	18.7	1.4	0.2	0	0	-
U2 - Beta(0.7, 0.3)	18.7	1.4	0.2	0	0	-
U2 - Beta(1, 1)	18.7	1.4	0.2	0	0	-
U3 - Beta(0.3, 0.7)	18.7	1.4	0.2	0	0	-
U3 - Beta(0.5, 0.5)	18.7	1.4	0.2	0	0	-
U3 - Beta(0.7, 0.3)	18.8	1.4	0.2	0	0	-
U3 - Beta(1, 1)	18.7	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.8	0.5	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.8	0.5	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.8	0.5	0.1	0	0	-
U2 - Beta(1, 1)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.8	0.5	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.8	0.5	0.1	0	0	-
U3 - Beta(1, 1)	3.8	0.5	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						880
U2 - Beta(0.5, 0.5)						880.5
U2 - Beta(0.7, 0.3)						879
U2 - Beta(1, 1)						880.7
U3 - Beta(0.3, 0.7)						880
U3 - Beta(0.5, 0.5)						880.4
U3 - Beta(0.7, 0.3)						882.3
U3 - Beta(1, 1)						880.5
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.6	51.6	5	0	0	11.8
U2 - Beta(0.5, 0.5)	30.6	51.3	5.8	0.1	0	12.2
U2 - Beta(0.7, 0.3)	30.3	51.8	6.1	0.1	0	11.7
U2 - Beta(1, 1)	32	50	6.1	0.1	0	11.8
U3 - Beta(0.3, 0.7)	32.2	51.5	4.3	0.1	0	11.9
U3 - Beta(0.5, 0.5)	32	50.7	5.4	0.1	0	11.8
U3 - Beta(0.7, 0.3)	33	49.4	5.5	0.1	0	12
U3 - Beta(1, 1)	34	48.1	5.6	0.1	0	12.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.4	13	4.8	0.7	0	-
U2 - Beta(0.5, 0.5)	9.6	12.8	4.8	0.7	0	-
U2 - Beta(0.7, 0.3)	9.6	12.8	4.8	0.7	0	-
U2 - Beta(1, 1)	9.8	12.7	4.7	0.7	0	-
U3 - Beta(0.3, 0.7)	9.5	12.9	4.8	0.7	0	-
U3 - Beta(0.5, 0.5)	9.6	12.8	4.7	0.7	0	-
U3 - Beta(0.7, 0.3)	9.6	12.8	4.7	0.7	0	-
U3 - Beta(1, 1)	9.7	12.7	4.7	0.7	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	5.6	1.4	0.3	0	-
U2 - Beta(0.5, 0.5)	2.7	5.6	1.4	0.3	0	-
U2 - Beta(0.7, 0.3)	2.7	5.6	1.4	0.3	0	-
U2 - Beta(1, 1)	2.8	5.5	1.4	0.3	0	-
U3 - Beta(0.3, 0.7)	2.6	5.6	1.4	0.3	0	-
U3 - Beta(0.5, 0.5)	2.7	5.6	1.4	0.3	0	-
U3 - Beta(0.7, 0.3)	2.7	5.6	1.4	0.3	0	-
U3 - Beta(1, 1)	2.7	5.6	1.4	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						982.1
U2 - Beta(0.5, 0.5)						980
U2 - Beta(0.7, 0.3)						979.6
U2 - Beta(1, 1)						978.7
U3 - Beta(0.3, 0.7)						981
U3 - Beta(0.5, 0.5)						980.9
U3 - Beta(0.7, 0.3)						979.1
U3 - Beta(1, 1)						978.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.5	1	0	0	0	93.5
U2 - Beta(0.5, 0.5)	5.7	1	0	0	0	93.3
U2 - Beta(0.7, 0.3)	5.8	1.1	0	0	0	93.1
U2 - Beta(1, 1)	5.7	1	0	0	0	93.3
U3 - Beta(0.3, 0.7)	5.6	1	0	0	0	93.4
U3 - Beta(0.5, 0.5)	5.7	1	0	0	0	93.3
U3 - Beta(0.7, 0.3)	5.5	1.1	0	0	0	93.4
U3 - Beta(1, 1)	5.8	1.1	0	0	0	93.1
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	1.4	0.2	0	0	-
U2 - Beta(0.5, 0.5)	11.8	1.4	0.2	0	0	-
U2 - Beta(0.7, 0.3)	11.8	1.4	0.2	0	0	-
U2 - Beta(1, 1)	11.7	1.4	0.2	0	0	-
U3 - Beta(0.3, 0.7)	11.8	1.4	0.2	0	0	-
U3 - Beta(0.5, 0.5)	11.8	1.4	0.2	0	0	-
U3 - Beta(0.7, 0.3)	11.8	1.4	0.2	0	0	-
U3 - Beta(1, 1)	11.8	1.4	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	0.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.5	0.9	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.5	0.9	0.1	0	0	-
U2 - Beta(1, 1)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.5	0.9	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.5	0.9	0.1	0	0	-
U3 - Beta(1, 1)	3.5	0.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						478.2
U2 - Beta(0.5, 0.5)						478.9
U2 - Beta(0.7, 0.3)						478.2
U2 - Beta(1, 1)						476.7
U3 - Beta(0.3, 0.7)						478
U3 - Beta(0.5, 0.5)						478.3
U3 - Beta(0.7, 0.3)						478.2
U3 - Beta(1, 1)						478.6
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	1.7	3.6	46	48.4	0.1
U2 - Beta(0.5, 0.5)	0.5	2.4	3.6	48.4	45	0.1
U2 - Beta(0.7, 0.3)	0.4	1.9	4	50	43.6	0.1
U2 - Beta(1, 1)	0.6	2.1	4.6	47.4	45.2	0.1
U3 - Beta(0.3, 0.7)	1.3	2.5	5.2	48.6	42.3	0.1
U3 - Beta(0.5, 0.5)	1.3	2.9	5.1	51.6	39	0.1
U3 - Beta(0.7, 0.3)	1.9	2.9	4.9	53.1	37	0.2
U3 - Beta(1, 1)	1.9	3.1	5.3	49.9	39.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.1	3.5	8.2	12	-
U2 - Beta(0.5, 0.5)	3.2	3.1	3.5	8.4	11.8	-
U2 - Beta(0.7, 0.3)	3.2	3.1	3.6	8.7	11.5	-
U2 - Beta(1, 1)	3.2	3.1	3.5	8.3	11.9	-
U3 - Beta(0.3, 0.7)	3.1	3.1	3.6	8.1	12	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.5	8.4	11.8	-
U3 - Beta(0.7, 0.3)	3.2	3.1	3.6	8.7	11.4	-
U3 - Beta(1, 1)	3.2	3.1	3.5	8.2	11.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.3	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.1	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.1	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102.3
U2 - Beta(0.5, 0.5)						1102.2
U2 - Beta(0.7, 0.3)						1101.6
U2 - Beta(1, 1)						1102.4
U3 - Beta(0.3, 0.7)						1102.3
U3 - Beta(0.5, 0.5)						1102.5
U3 - Beta(0.7, 0.3)						1100.3
U3 - Beta(1, 1)						1102

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	36.6	37.4	22.5	1.8	0	1.7
U2 - Beta(0.5, 0.5)	37	36.8	22.7	1.7	0	1.8
U2 - Beta(0.7, 0.3)	36	37.6	22.8	1.8	0	1.8
U2 - Beta(1, 1)	37.5	35.5	23.3	1.9	0	1.8
U3 - Beta(0.3, 0.7)	39.9	38.1	18.8	1.4	0	1.8
U3 - Beta(0.5, 0.5)	40	37.2	19.6	1.5	0	1.7
U3 - Beta(0.7, 0.3)	42.4	35.8	18.4	1.6	0	1.8
U3 - Beta(1, 1)	42.5	35.6	18.8	1.3	0	1.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	9.3	8.2	1.6	0.2	-
U2 - Beta(0.5, 0.5)	10.6	9.4	8	1.7	0.2	-
U2 - Beta(0.7, 0.3)	10.6	9.5	7.8	1.6	0.2	-
U2 - Beta(1, 1)	10.7	9.4	7.8	1.6	0.2	-
U3 - Beta(0.3, 0.7)	10.6	9.2	8.1	1.6	0.2	-
U3 - Beta(0.5, 0.5)	10.6	9.4	7.9	1.6	0.2	-
U3 - Beta(0.7, 0.3)	10.8	9.2	7.9	1.6	0.2	-
U3 - Beta(1, 1)	10.7	9.3	8	1.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	4	2.3	0.6	0	-
U2 - Beta(0.5, 0.5)	3	4.1	2.2	0.6	0	-
U2 - Beta(0.7, 0.3)	3.1	4.1	2.2	0.6	0	-
U2 - Beta(1, 1)	3.1	4.1	2.2	0.6	0	-
U3 - Beta(0.3, 0.7)	3	4	2.2	0.6	0	-
U3 - Beta(0.5, 0.5)	3.1	4.1	2.2	0.6	0	-
U3 - Beta(0.7, 0.3)	3.1	4	2.2	0.6	0	-
U3 - Beta(1, 1)	3.1	4.1	2.2	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1091
U2 - Beta(0.5, 0.5)						1089.5
U2 - Beta(0.7, 0.3)						1088
U2 - Beta(1, 1)						1089
U3 - Beta(0.3, 0.7)						1090.3
U3 - Beta(0.5, 0.5)						1089.4
U3 - Beta(0.7, 0.3)						1089.2
U3 - Beta(1, 1)						1089.8
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31	29.7	17.8	1.7	0	19.8
U2 - Beta(0.5, 0.5)	30.8	29.2	17.6	2.1	0	20.3
U2 - Beta(0.7, 0.3)	31.4	29.3	17	2.4	0	19.9
U2 - Beta(1, 1)	31.5	29.5	16.8	2.3	0	19.9
U3 - Beta(0.3, 0.7)	31.6	30.2	16.6	1.7	0	19.9
U3 - Beta(0.5, 0.5)	34.1	28.9	15.5	1.6	0	19.9
U3 - Beta(0.7, 0.3)	34.6	29.9	13.4	2.2	0	19.9
U3 - Beta(1, 1)	34.6	29.3	14.5	1.6	0	20
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	7.2	6.3	1.5	0.1	-
U2 - Beta(0.5, 0.5)	11.8	7.2	6.3	1.5	0.1	-
U2 - Beta(0.7, 0.3)	11.8	7.3	6.2	1.5	0.1	-
U2 - Beta(1, 1)	11.8	7.3	6.2	1.5	0.1	-
U3 - Beta(0.3, 0.7)	11.7	7.2	6.4	1.5	0.1	-
U3 - Beta(0.5, 0.5)	11.8	7.3	6.2	1.5	0.1	-
U3 - Beta(0.7, 0.3)	11.9	7.4	6	1.5	0.1	-
U3 - Beta(1, 1)	11.8	7.4	6.2	1.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	3.4	1.9	0.6	0	-
U2 - Beta(0.5, 0.5)	3.7	3.4	1.9	0.6	0	-
U2 - Beta(0.7, 0.3)	3.7	3.5	1.9	0.6	0	-
U2 - Beta(1, 1)	3.7	3.4	1.9	0.6	0	-
U3 - Beta(0.3, 0.7)	3.6	3.4	1.9	0.6	0	-
U3 - Beta(0.5, 0.5)	3.7	3.5	1.9	0.6	0	-
U3 - Beta(0.7, 0.3)	3.7	3.5	1.9	0.5	0	-
U3 - Beta(1, 1)	3.6	3.5	1.9	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						951.6
U2 - Beta(0.5, 0.5)						951.3
U2 - Beta(0.7, 0.3)						950.4
U2 - Beta(1, 1)						951.2
U3 - Beta(0.3, 0.7)						952.7
U3 - Beta(0.5, 0.5)						950.8
U3 - Beta(0.7, 0.3)						950.6
U3 - Beta(1, 1)						951.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.8	30.7	32.5	3	0	12
U2 - Beta(0.5, 0.5)	22.3	30.5	32.1	3.2	0	11.9
U2 - Beta(0.7, 0.3)	23.5	29.9	31.7	2.9	0.1	11.9
U2 - Beta(1, 1)	22.3	30.1	32.2	3.5	0	11.9
U3 - Beta(0.3, 0.7)	23.5	30.9	31	2.7	0	11.9
U3 - Beta(0.5, 0.5)	25.1	29.9	29.5	3.5	0	12
U3 - Beta(0.7, 0.3)	27	29.7	27.9	3.5	0	11.9
U3 - Beta(1, 1)	27.3	30.5	27.7	2.7	0	11.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	7.3	8.9	2.6	0.5	-
U2 - Beta(0.5, 0.5)	8.9	7.5	8.5	2.6	0.5	-
U2 - Beta(0.7, 0.3)	9	7.5	8.6	2.6	0.5	-
U2 - Beta(1, 1)	8.9	7.5	8.7	2.6	0.5	-
U3 - Beta(0.3, 0.7)	8.8	7.3	8.8	2.6	0.5	-
U3 - Beta(0.5, 0.5)	8.9	7.5	8.6	2.6	0.5	-
U3 - Beta(0.7, 0.3)	8.9	7.5	8.6	2.6	0.5	-
U3 - Beta(1, 1)	9	7.4	8.6	2.6	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.6	2.6	0.8	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.7	2.5	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.6	2.5	0.8	0.1	-
U2 - Beta(1, 1)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.6	2.6	0.8	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.6	2.5	0.8	0.1	-
U3 - Beta(1, 1)	2.5	3.6	2.5	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1020
U2 - Beta(0.5, 0.5)						1018.6
U2 - Beta(0.7, 0.3)						1019.1
U2 - Beta(1, 1)						1018.2
U3 - Beta(0.3, 0.7)						1020.5
U3 - Beta(0.5, 0.5)						1018.3
U3 - Beta(0.7, 0.3)						1018.6
U3 - Beta(1, 1)						1019.3
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.6	21.6	0.6	0	0	57.2
U2 - Beta(0.5, 0.5)	20.3	21.7	0.6	0	0	57.4
U2 - Beta(0.7, 0.3)	20.1	22	0.7	0	0	57.2
U2 - Beta(1, 1)	19.9	22.1	0.5	0	0	57.5
U3 - Beta(0.3, 0.7)	20.7	21.4	0.6	0	0	57.3
U3 - Beta(0.5, 0.5)	20.8	21.4	0.5	0	0	57.3
U3 - Beta(0.7, 0.3)	19.9	21.8	0.8	0	0	57.5
U3 - Beta(1, 1)	20	22.1	0.6	0	0	57.3
Mean administrations						
U2 - Beta(0.3, 0.7)	10.9	6.9	2.4	0.3	0	-
U2 - Beta(0.5, 0.5)	10.9	6.9	2.3	0.3	0	-
U2 - Beta(0.7, 0.3)	10.9	6.9	2.3	0.3	0	-
U2 - Beta(1, 1)	11	6.8	2.3	0.3	0	-
U3 - Beta(0.3, 0.7)	10.9	6.8	2.4	0.3	0	-
U3 - Beta(0.5, 0.5)	10.9	6.9	2.3	0.3	0	-
U3 - Beta(0.7, 0.3)	10.9	6.8	2.3	0.3	0	-
U3 - Beta(1, 1)	10.9	6.9	2.4	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	3.2	3	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	3.2	3	0.7	0.1	0	-
U2 - Beta(1, 1)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	3.2	3	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	3.2	3	0.7	0.1	0	-
U3 - Beta(1, 1)	3.2	3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						733.4
U2 - Beta(0.5, 0.5)						730.5
U2 - Beta(0.7, 0.3)						731.5
U2 - Beta(1, 1)						731.5
U3 - Beta(0.3, 0.7)						731.6
U3 - Beta(0.5, 0.5)						732.4
U3 - Beta(0.7, 0.3)						730.3
U3 - Beta(1, 1)						731.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	77	17.9	1.3	0	0	3.8
U2 - Beta(0.5, 0.5)	77	18.2	1	0	0	3.8
U2 - Beta(0.7, 0.3)	76	19.1	1.2	0	0	3.7
U2 - Beta(1, 1)	77	17.9	1.4	0	0	3.7
U3 - Beta(0.3, 0.7)	77.9	17.4	0.8	0	0	3.9
U3 - Beta(0.5, 0.5)	78.2	16.8	1.2	0	0	3.8
U3 - Beta(0.7, 0.3)	77	18.3	1.1	0	0	3.6
U3 - Beta(1, 1)	78.2	17.1	0.9	0	0	3.8
Mean administrations						
U2 - Beta(0.3, 0.7)	20	6.8	2.3	0.2	0	-
U2 - Beta(0.5, 0.5)	20.1	6.7	2.3	0.2	0	-
U2 - Beta(0.7, 0.3)	20.2	6.6	2.3	0.2	0	-
U2 - Beta(1, 1)	20.1	6.7	2.3	0.2	0	-
U3 - Beta(0.3, 0.7)	19.9	6.8	2.4	0.2	0	-
U3 - Beta(0.5, 0.5)	20.1	6.6	2.3	0.2	0	-
U3 - Beta(0.7, 0.3)	20.2	6.6	2.3	0.2	0	-
U3 - Beta(1, 1)	20.1	6.7	2.3	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.2	2.3	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	5.3	2.3	0.6	0.1	0	-
U2 - Beta(0.7, 0.3)	5.3	2.3	0.7	0.1	0	-
U2 - Beta(1, 1)	5.3	2.3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	5.2	2.3	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	5.3	2.3	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	5.3	2.3	0.6	0.1	0	-
U3 - Beta(1, 1)	5.3	2.3	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1159.4
U2 - Beta(0.5, 0.5)						1158.1
U2 - Beta(0.7, 0.3)						1158.7
U2 - Beta(1, 1)						1158.8
U3 - Beta(0.3, 0.7)						1159
U3 - Beta(0.5, 0.5)						1158.1
U3 - Beta(0.7, 0.3)						1158.9
U3 - Beta(1, 1)						1158.5
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.3	19	21.4	35.5	4.2	1.6
U2 - Beta(0.5, 0.5)	18.7	19.3	21	35	4.4	1.6
U2 - Beta(0.7, 0.3)	19.5	20.4	19.5	33.7	5.3	1.6
U2 - Beta(1, 1)	19.9	19.9	20.6	33.3	4.8	1.5
U3 - Beta(0.3, 0.7)	22.6	22.7	17.6	32.6	2.9	1.6
U3 - Beta(0.5, 0.5)	24.5	22.4	17.1	31	3.4	1.6
U3 - Beta(0.7, 0.3)	25.2	23.6	15.6	30.6	3.4	1.6
U3 - Beta(1, 1)	27.8	22.6	15.6	29.1	3.3	1.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.4	5.3	6.8	8.2	4.1	-
U2 - Beta(0.5, 0.5)	5.4	5.4	6.8	8	4.2	-
U2 - Beta(0.7, 0.3)	5.5	5.5	6.7	8	4.1	-
U2 - Beta(1, 1)	5.4	5.5	6.7	8	4.2	-
U3 - Beta(0.3, 0.7)	5.4	5.2	6.8	8.2	4.1	-
U3 - Beta(0.5, 0.5)	5.4	5.3	6.7	8.1	4.2	-
U3 - Beta(0.7, 0.3)	5.4	5.4	6.8	8.1	4.1	-
U3 - Beta(1, 1)	5.5	5.4	6.7	7.9	4.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.1	2.2	2.9	1.1	-
U2 - Beta(0.5, 0.5)	1.1	3.1	2.2	2.8	1.1	-
U2 - Beta(0.7, 0.3)	1.1	3.1	2.2	2.8	1.1	-
U2 - Beta(1, 1)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(0.3, 0.7)	1.1	3.1	2.2	2.9	1.1	-
U3 - Beta(0.5, 0.5)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(0.7, 0.3)	1.1	3.1	2.2	2.8	1.1	-
U3 - Beta(1, 1)	1.1	3.1	2.2	2.8	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1071.2
U2 - Beta(0.5, 0.5)						1072.4
U2 - Beta(0.7, 0.3)						1071.3
U2 - Beta(1, 1)						1071.7
U3 - Beta(0.3, 0.7)						1072.8
U3 - Beta(0.5, 0.5)						1072.7
U3 - Beta(0.7, 0.3)						1071.9
U3 - Beta(1, 1)						1071.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	35.3	30.4	26.3	3.3	0.1
U2 - Beta(0.5, 0.5)	5.3	37.6	29.3	23.3	4.4	0.1
U2 - Beta(0.7, 0.3)	5.3	40.8	26.8	22.3	4.7	0.1
U2 - Beta(1, 1)	5.5	37.6	30.9	22.6	3.3	0.1
U3 - Beta(0.3, 0.7)	6.2	43.1	27.8	20.1	2.7	0.1
U3 - Beta(0.5, 0.5)	7.2	46	25.2	18.9	2.6	0.1
U3 - Beta(0.7, 0.3)	6.8	47.5	25	17.5	3.1	0.1
U3 - Beta(1, 1)	6.6	46.2	26.2	18.5	2.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	5.9	6.5	8.2	5.2	-
U2 - Beta(0.5, 0.5)	4.2	5.9	6.7	8.1	5.1	-
U2 - Beta(0.7, 0.3)	4.2	6	6.7	8	5	-
U2 - Beta(1, 1)	4.2	5.8	6.6	8.2	5.1	-
U3 - Beta(0.3, 0.7)	4.2	5.9	6.7	8.1	5.2	-
U3 - Beta(0.5, 0.5)	4.2	5.9	6.7	8.2	5.1	-
U3 - Beta(0.7, 0.3)	4.3	6	6.7	8	5	-
U3 - Beta(1, 1)	4.2	5.9	6.6	8.1	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2.1	1.4	-
U2 - Beta(1, 1)	0.5	2.9	1.7	2.1	1.3	-
U3 - Beta(0.3, 0.7)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(0.5, 0.5)	0.5	2.9	1.7	2.1	1.4	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2.1	1.3	-
U3 - Beta(1, 1)	0.5	2.9	1.7	2.1	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1181.4
U2 - Beta(0.5, 0.5)						1181.4
U2 - Beta(0.7, 0.3)						1180.3
U2 - Beta(1, 1)						1181.4
U3 - Beta(0.3, 0.7)						1181
U3 - Beta(0.5, 0.5)						1181.7
U3 - Beta(0.7, 0.3)						1179.1
U3 - Beta(1, 1)						1181.2
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.9	13.9	34.5	31.7	9.3	0.7
U2 - Beta(0.5, 0.5)	9.5	13.9	34.2	33	8.7	0.7
U2 - Beta(0.7, 0.3)	9.5	16.2	32.9	33	7.7	0.7
U2 - Beta(1, 1)	9.5	15.2	35.8	31.1	7.7	0.7
U3 - Beta(0.3, 0.7)	13	19.2	34.4	26	6.6	0.8
U3 - Beta(0.5, 0.5)	14.4	19	33.6	27.4	4.9	0.7
U3 - Beta(0.7, 0.3)	15.2	20.2	32.3	26.8	4.8	0.7
U3 - Beta(1, 1)	15.8	19.4	32.8	26.9	4.4	0.7
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	4.4	7.8	7.9	6.1	-
U2 - Beta(0.5, 0.5)	3.8	4.5	7.8	8	5.8	-
U2 - Beta(0.7, 0.3)	3.9	4.6	7.8	7.9	5.7	-
U2 - Beta(1, 1)	3.8	4.5	7.8	8	5.8	-
U3 - Beta(0.3, 0.7)	3.8	4.3	7.9	7.8	6.1	-
U3 - Beta(0.5, 0.5)	3.8	4.5	7.8	8	5.8	-
U3 - Beta(0.7, 0.3)	3.9	4.6	7.8	7.9	5.8	-
U3 - Beta(1, 1)	3.9	4.5	7.8	8	5.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.9	2.5	2.9	2.1	-
U2 - Beta(0.5, 0.5)	0.4	2.9	2.5	3	2	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	1.9	-
U2 - Beta(1, 1)	0.4	2.9	2.5	3	2	-
U3 - Beta(0.3, 0.7)	0.3	2.9	2.5	2.9	2.1	-
U3 - Beta(0.5, 0.5)	0.3	2.9	2.5	2.9	2	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	2	-
U3 - Beta(1, 1)	0.4	2.9	2.5	2.9	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057
U2 - Beta(0.5, 0.5)						1055.8
U2 - Beta(0.7, 0.3)						1055.6
U2 - Beta(1, 1)						1055.3
U3 - Beta(0.3, 0.7)						1057.6
U3 - Beta(0.5, 0.5)						1056.3
U3 - Beta(0.7, 0.3)						1055
U3 - Beta(1, 1)						1055.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	9.5	19.1	44.8	18	0.1
U2 - Beta(0.5, 0.5)	9.6	9.7	16.9	46.1	17.6	0.1
U2 - Beta(0.7, 0.3)	10	10.6	15.9	46	17.4	0.1
U2 - Beta(1, 1)	10.9	10.8	15.9	44.4	17.9	0.1
U3 - Beta(0.3, 0.7)	11.8	11.5	17.9	42.4	16.2	0.2
U3 - Beta(0.5, 0.5)	13.2	11.6	16.2	43.9	14.9	0.2
U3 - Beta(0.7, 0.3)	15.2	12.6	15.3	42.4	14.4	0.1
U3 - Beta(1, 1)	15.3	13.2	15.6	41.4	14.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.6	5.6	8.4	9	-
U2 - Beta(0.5, 0.5)	3.5	3.7	5.6	8.4	8.8	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.6	8.5	8.7	-
U2 - Beta(1, 1)	3.5	3.7	5.6	8.2	9	-
U3 - Beta(0.3, 0.7)	3.4	3.6	5.6	8.3	9	-
U3 - Beta(0.5, 0.5)	3.5	3.7	5.6	8.4	8.8	-
U3 - Beta(0.7, 0.3)	3.5	3.8	5.6	8.4	8.7	-
U3 - Beta(1, 1)	3.5	3.7	5.6	8.3	9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.7	3.1	2.3	-
U2 - Beta(0.5, 0.5)	0.2	2.7	1.7	3.1	2.3	-
U2 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.2	-
U2 - Beta(1, 1)	0.2	2.7	1.7	3	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.7	3.1	2.3	-
U3 - Beta(0.5, 0.5)	0.2	2.7	1.7	3.1	2.2	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.2	-
U3 - Beta(1, 1)	0.2	2.7	1.7	3.1	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1103
U2 - Beta(0.5, 0.5)						1100.2
U2 - Beta(0.7, 0.3)						1100.2
U2 - Beta(1, 1)						1101.8
U3 - Beta(0.3, 0.7)						1101.8
U3 - Beta(0.5, 0.5)						1100.7
U3 - Beta(0.7, 0.3)						1100
U3 - Beta(1, 1)						1100.3
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.5	20.3	46.3	10.6	1.9	6.4
U2 - Beta(0.5, 0.5)	16	20	45.7	10.8	1.5	6
U2 - Beta(0.7, 0.3)	17.3	18.4	45.7	10.7	1.8	6.1
U2 - Beta(1, 1)	18.4	17.5	46.1	9.8	1.8	6.4
U3 - Beta(0.3, 0.7)	18.8	21.1	42.5	9.7	1.5	6.4
U3 - Beta(0.5, 0.5)	21.1	19.6	42.7	9	1.3	6.3
U3 - Beta(0.7, 0.3)	23.9	19.3	41.4	8.2	1.2	6
U3 - Beta(1, 1)	25.2	18.8	40.3	8.4	1.3	6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	6.6	10.4	4.5	2	-
U2 - Beta(0.5, 0.5)	5.5	6.8	10.3	4.5	1.9	-
U2 - Beta(0.7, 0.3)	5.6	6.8	10.1	4.5	1.9	-
U2 - Beta(1, 1)	5.6	6.7	10.2	4.5	2	-
U3 - Beta(0.3, 0.7)	5.4	6.6	10.4	4.5	2	-
U3 - Beta(0.5, 0.5)	5.5	6.7	10.4	4.5	1.9	-
U3 - Beta(0.7, 0.3)	5.6	6.7	10.2	4.4	1.9	-
U3 - Beta(1, 1)	5.7	6.7	10.2	4.5	1.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.6	3.3	1.6	0.7	-
U2 - Beta(0.5, 0.5)	1.1	3.7	3.3	1.6	0.6	-
U2 - Beta(0.7, 0.3)	1.2	3.7	3.3	1.6	0.6	-
U2 - Beta(1, 1)	1.2	3.6	3.3	1.6	0.6	-
U3 - Beta(0.3, 0.7)	1.1	3.6	3.3	1.6	0.6	-
U3 - Beta(0.5, 0.5)	1.1	3.6	3.3	1.6	0.6	-
U3 - Beta(0.7, 0.3)	1.2	3.7	3.3	1.6	0.6	-
U3 - Beta(1, 1)	1.2	3.6	3.3	1.6	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022.3
U2 - Beta(0.5, 0.5)						1021.2
U2 - Beta(0.7, 0.3)						1021.1
U2 - Beta(1, 1)						1020.7
U3 - Beta(0.3, 0.7)						1023.5
U3 - Beta(0.5, 0.5)						1022.4
U3 - Beta(0.7, 0.3)						1021.1
U3 - Beta(1, 1)						1020.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.4	50.1	31.1	8.2	1.1	0.1
U2 - Beta(0.5, 0.5)	10.1	49.5	29.8	9.6	0.9	0.1
U2 - Beta(0.7, 0.3)	8.9	50.7	30.3	9.1	0.9	0.1
U2 - Beta(1, 1)	10.4	50.4	30.8	7.6	0.7	0.1
U3 - Beta(0.3, 0.7)	11.5	53.6	27.5	6.9	0.4	0.1
U3 - Beta(0.5, 0.5)	12.1	53	26.5	7.5	0.8	0.1
U3 - Beta(0.7, 0.3)	12.2	52.9	25.8	8.3	0.7	0.1
U3 - Beta(1, 1)	13.1	56.2	23.4	6.7	0.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	9.5	9.6	4.5	1.9	-
U2 - Beta(0.5, 0.5)	4.6	9.7	9.4	4.5	1.8	-
U2 - Beta(0.7, 0.3)	4.7	9.8	9.3	4.4	1.8	-
U2 - Beta(1, 1)	4.7	9.6	9.4	4.4	1.9	-
U3 - Beta(0.3, 0.7)	4.5	9.5	9.6	4.5	1.8	-
U3 - Beta(0.5, 0.5)	4.6	9.6	9.5	4.5	1.8	-
U3 - Beta(0.7, 0.3)	4.7	9.8	9.4	4.4	1.8	-
U3 - Beta(1, 1)	4.6	9.7	9.4	4.5	1.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	4	2.4	1.3	0.5	-
U2 - Beta(0.5, 0.5)	0.7	4.1	2.3	1.4	0.5	-
U2 - Beta(0.7, 0.3)	0.7	4.1	2.3	1.3	0.5	-
U2 - Beta(1, 1)	0.7	4	2.3	1.3	0.6	-
U3 - Beta(0.3, 0.7)	0.6	4	2.4	1.4	0.5	-
U3 - Beta(0.5, 0.5)	0.7	4	2.3	1.4	0.5	-
U3 - Beta(0.7, 0.3)	0.7	4.1	2.3	1.3	0.5	-
U3 - Beta(1, 1)	0.7	4.1	2.3	1.4	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1162.6
U2 - Beta(0.5, 0.5)						1161
U2 - Beta(0.7, 0.3)						1159.7
U2 - Beta(1, 1)						1162.2
U3 - Beta(0.3, 0.7)						1161.6
U3 - Beta(0.5, 0.5)						1160.8
U3 - Beta(0.7, 0.3)						1160.4
U3 - Beta(1, 1)						1161.6

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.4	35.6	8.1	1.4	0.5	33
U2 - Beta(0.5, 0.5)	20.5	36.9	7.9	1.5	0.4	32.8
U2 - Beta(0.7, 0.3)	21.1	36.2	7.9	1.4	0.4	33
U2 - Beta(1, 1)	22.7	35	7.4	1.3	0.4	33.2
U3 - Beta(0.3, 0.7)	23.2	35.3	6.7	1.4	0.3	33.1
U3 - Beta(0.5, 0.5)	24.8	33.7	6.8	1.1	0.4	33.2
U3 - Beta(0.7, 0.3)	25.2	34.2	5.9	1.3	0.2	33.2
U3 - Beta(1, 1)	25.6	33.8	5.8	1.4	0.3	33.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.5	8.4	4.5	1.4	0.4	-
U2 - Beta(0.5, 0.5)	9.5	8.5	4.5	1.4	0.4	-
U2 - Beta(0.7, 0.3)	9.6	8.4	4.5	1.4	0.4	-
U2 - Beta(1, 1)	9.7	8.4	4.4	1.3	0.4	-
U3 - Beta(0.3, 0.7)	9.5	8.4	4.6	1.4	0.4	-
U3 - Beta(0.5, 0.5)	9.6	8.4	4.5	1.4	0.4	-
U3 - Beta(0.7, 0.3)	9.6	8.5	4.4	1.4	0.4	-
U3 - Beta(1, 1)	9.7	8.4	4.4	1.4	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	4	1.5	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.7	4	1.5	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.7	4	1.4	0.6	0.1	-
U2 - Beta(1, 1)	2.7	4	1.4	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.7	4	1.5	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.7	4	1.5	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.7	4	1.4	0.6	0.1	-
U3 - Beta(1, 1)	2.7	4	1.4	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						846.1
U2 - Beta(0.5, 0.5)						846.3
U2 - Beta(0.7, 0.3)						846.2
U2 - Beta(1, 1)						846.1
U3 - Beta(0.3, 0.7)						846.1
U3 - Beta(0.5, 0.5)						845.5
U3 - Beta(0.7, 0.3)						846.5
U3 - Beta(1, 1)						845.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.9	43.9	6.5	0	0	21.7
U2 - Beta(0.5, 0.5)	28.8	42.5	6.8	0	0.1	21.8
U2 - Beta(0.7, 0.3)	28.3	42.2	6.4	0.2	0.1	22.8
U2 - Beta(1, 1)	29.1	42.7	5.9	0.2	0	22.1
U3 - Beta(0.3, 0.7)	30.6	42.5	4.9	0.1	0	21.9
U3 - Beta(0.5, 0.5)	30.1	42.6	5.3	0	0	22
U3 - Beta(0.7, 0.3)	31.1	41.9	5	0.3	0	21.7
U3 - Beta(1, 1)	32.2	41.6	3.8	0.4	0	22
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.9	5	1.2	0.2	-
U2 - Beta(0.5, 0.5)	8.9	10.9	5	1.1	0.2	-
U2 - Beta(0.7, 0.3)	9	10.9	4.9	1.1	0.2	-
U2 - Beta(1, 1)	9	10.9	4.8	1.2	0.2	-
U3 - Beta(0.3, 0.7)	8.8	11.1	4.9	1.2	0.2	-
U3 - Beta(0.5, 0.5)	8.9	11	4.9	1.1	0.2	-
U3 - Beta(0.7, 0.3)	8.9	11.1	4.9	1.1	0.2	-
U3 - Beta(1, 1)	9.1	10.9	4.8	1.2	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	5.1	1.6	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.4	5.1	1.6	0.5	0.1	-
U2 - Beta(0.7, 0.3)	2.4	5.1	1.6	0.5	0.1	-
U2 - Beta(1, 1)	2.4	5.1	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.3	5.1	1.6	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.4	5.1	1.6	0.5	0.1	-
U3 - Beta(0.7, 0.3)	2.4	5.1	1.5	0.5	0.1	-
U3 - Beta(1, 1)	2.4	5.1	1.5	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						908.9
U2 - Beta(0.5, 0.5)						908.8
U2 - Beta(0.7, 0.3)						906.9
U2 - Beta(1, 1)						906.5
U3 - Beta(0.3, 0.7)						908
U3 - Beta(0.5, 0.5)						907.2
U3 - Beta(0.7, 0.3)						909.3
U3 - Beta(1, 1)						907.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	2.4	0.3	0	0	86.6
U2 - Beta(0.5, 0.5)	12.3	2.4	0.3	0	0	85
U2 - Beta(0.7, 0.3)	12.5	2.5	0.5	0	0	84.5
U2 - Beta(1, 1)	12	2.7	0.4	0	0	84.9
U3 - Beta(0.3, 0.7)	10.5	2.4	0.3	0	0	86.8
U3 - Beta(0.5, 0.5)	12.3	2.5	0.4	0	0	84.8
U3 - Beta(0.7, 0.3)	11.8	2.6	0.3	0	0	85.3
U3 - Beta(1, 1)	12.1	2.4	0.3	0	0	85.2
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	4.1	0.9	0.3	0	-
U2 - Beta(0.5, 0.5)	11.6	4.3	1	0.4	0	-
U2 - Beta(0.7, 0.3)	11.6	4.3	1	0.4	0.1	-
U2 - Beta(1, 1)	11.6	4.3	1	0.4	0	-
U3 - Beta(0.3, 0.7)	11.8	4	0.8	0.3	0	-
U3 - Beta(0.5, 0.5)	11.7	4.2	1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	11.6	4.3	1	0.4	0.1	-
U3 - Beta(1, 1)	11.7	4.3	1	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	2.3	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3.2	2.4	0.4	0.2	0	-
U2 - Beta(0.7, 0.3)	3.3	2.4	0.3	0.2	0	-
U2 - Beta(1, 1)	3.2	2.4	0.4	0.2	0	-
U3 - Beta(0.3, 0.7)	3.4	2.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	3.3	2.4	0.4	0.2	0	-
U3 - Beta(0.7, 0.3)	3.3	2.4	0.3	0.2	0	-
U3 - Beta(1, 1)	3.3	2.4	0.3	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						593.9
U2 - Beta(0.5, 0.5)						606.4
U2 - Beta(0.7, 0.3)						610.4
U2 - Beta(1, 1)						607.2
U3 - Beta(0.3, 0.7)						593.1
U3 - Beta(0.5, 0.5)						607.8
U3 - Beta(0.7, 0.3)						606.2
U3 - Beta(1, 1)						606.8
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	39	37.4	19.4	0.9	0	3.3
U2 - Beta(0.5, 0.5)	33.8	27.2	31.2	4.8	0.3	2.7
U2 - Beta(0.7, 0.3)	33.9	28.3	28.4	6.3	0.6	2.5
U2 - Beta(1, 1)	31.4	28.7	32.5	4.9	0.2	2.3
U3 - Beta(0.3, 0.7)	46.3	37.9	11.4	0.4	0	4
U3 - Beta(0.5, 0.5)	38.1	27.2	28	4	0.1	2.6
U3 - Beta(0.7, 0.3)	36.6	26.9	27.2	6.3	0.5	2.5
U3 - Beta(1, 1)	36.1	27.9	29.2	4	0.2	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	15.1	10.2	4.1	0.2	0	-
U2 - Beta(0.5, 0.5)	11.6	8.9	7.3	1.7	0.1	-
U2 - Beta(0.7, 0.3)	11.3	8.5	6.6	2.8	0.5	-
U2 - Beta(1, 1)	11.4	9.1	7.5	1.6	0.1	-
U3 - Beta(0.3, 0.7)	16.3	10.3	2.7	0.1	0	-
U3 - Beta(0.5, 0.5)	12.2	9.1	6.9	1.4	0	-
U3 - Beta(0.7, 0.3)	11.6	8.6	6.7	2.5	0.2	-
U3 - Beta(1, 1)	12.1	9.3	6.9	1.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	4.4	1.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3.1	3.9	2.2	0.6	0	-
U2 - Beta(0.7, 0.3)	3	3.7	1.9	0.8	0.2	-
U2 - Beta(1, 1)	3	4	2.2	0.5	0	-
U3 - Beta(0.3, 0.7)	5.1	4.5	0.9	0.1	0	-
U3 - Beta(0.5, 0.5)	3.3	4	2.1	0.5	0	-
U3 - Beta(0.7, 0.3)	3.1	3.8	1.9	0.8	0.1	-
U3 - Beta(1, 1)	3.3	4.1	2	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1054.2
U2 - Beta(0.5, 0.5)						1094.7
U2 - Beta(0.7, 0.3)						1102.4
U2 - Beta(1, 1)						1094.8
U3 - Beta(0.3, 0.7)						1045
U3 - Beta(0.5, 0.5)						1088.3
U3 - Beta(0.7, 0.3)						1097.4
U3 - Beta(1, 1)						1089.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.4	67.7	8.3	0.2	0	12.4
U2 - Beta(0.5, 0.5)	12.1	53.5	19.2	3.9	0.1	11.2
U2 - Beta(0.7, 0.3)	12.5	52.7	19	4.6	0.3	10.9
U2 - Beta(1, 1)	11.3	54.6	19.4	3.7	0.3	10.7
U3 - Beta(0.3, 0.7)	13.3	67.7	6.6	0.1	0	12.3
U3 - Beta(0.5, 0.5)	13	56.9	15.8	3.1	0	11.2
U3 - Beta(0.7, 0.3)	14	54	17.1	4	0.2	10.7
U3 - Beta(1, 1)	12.2	58.3	15.6	2.6	0	11.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.3	16.7	2.2	0.2	0	-
U2 - Beta(0.5, 0.5)	8.4	13	5.3	1.7	0.1	-
U2 - Beta(0.7, 0.3)	8.3	11.2	5.7	2.8	0.6	-
U2 - Beta(1, 1)	8.3	13.1	5.4	1.7	0.1	-
U3 - Beta(0.3, 0.7)	9.9	16.6	1.7	0.1	0	-
U3 - Beta(0.5, 0.5)	8.6	13.6	4.7	1.5	0.1	-
U3 - Beta(0.7, 0.3)	8.4	11.9	5.4	2.5	0.3	-
U3 - Beta(1, 1)	8.6	13.7	4.7	1.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	6.6	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.9	5.3	1.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.7	1.7	1	0.2	-
U2 - Beta(1, 1)	1.8	5.4	1.6	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.6	6.5	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	1.9	5.6	1.4	0.6	0	-
U3 - Beta(0.7, 0.3)	1.9	5	1.6	0.9	0.1	-
U3 - Beta(1, 1)	1.9	5.6	1.4	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1026.2
U2 - Beta(0.5, 0.5)						1044.1
U2 - Beta(0.7, 0.3)						1049.8
U2 - Beta(1, 1)						1045.4
U3 - Beta(0.3, 0.7)						1022.3
U3 - Beta(0.5, 0.5)						1040.7
U3 - Beta(0.7, 0.3)						1044.5
U3 - Beta(1, 1)						1040.8
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.5	9.2	44.3	39.1	0.8	2.1
U2 - Beta(0.5, 0.5)	6.6	9.9	26.3	50	5.4	1.8
U2 - Beta(0.7, 0.3)	6.8	10.8	25.9	46.5	8.2	1.8
U2 - Beta(1, 1)	6.4	9.2	25.2	51.7	5.4	2.1
U3 - Beta(0.3, 0.7)	6.5	13.3	40.5	37	0.5	2.2
U3 - Beta(0.5, 0.5)	8.8	11.4	23.6	50.3	3.9	2
U3 - Beta(0.7, 0.3)	9.4	11.7	24.2	46.3	6.4	2
U3 - Beta(1, 1)	10.5	11.6	21.5	50.8	3.5	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	6.4	11.5	7.1	0.2	-
U2 - Beta(0.5, 0.5)	4.4	5.3	8.2	10.2	1.7	-
U2 - Beta(0.7, 0.3)	4.6	5.4	7.5	9.1	3.2	-
U2 - Beta(1, 1)	4.5	5.4	8.1	10.1	1.7	-
U3 - Beta(0.3, 0.7)	5	6.9	11.2	6.6	0.1	-
U3 - Beta(0.5, 0.5)	4.7	5.5	8.2	10	1.4	-
U3 - Beta(0.7, 0.3)	4.8	5.7	7.7	9.3	2.3	-
U3 - Beta(1, 1)	4.9	5.7	7.9	9.9	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.5	4.1	2.4	0.1	-
U2 - Beta(0.5, 0.5)	0.6	3.3	2.4	3.3	0.6	-
U2 - Beta(0.7, 0.3)	0.7	3.4	2.2	3	0.9	-
U2 - Beta(1, 1)	0.6	3.4	2.4	3.3	0.6	-
U3 - Beta(0.3, 0.7)	0.9	3.8	4.1	2.3	0	-
U3 - Beta(0.5, 0.5)	0.7	3.4	2.5	3.3	0.5	-
U3 - Beta(0.7, 0.3)	0.8	3.5	2.3	3	0.7	-
U3 - Beta(1, 1)	0.8	3.5	2.3	3.2	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1037.2
U2 - Beta(0.5, 0.5)						1075.4
U2 - Beta(0.7, 0.3)						1082.3
U2 - Beta(1, 1)						1074.8
U3 - Beta(0.3, 0.7)						1028.9
U3 - Beta(0.5, 0.5)						1070.1
U3 - Beta(0.7, 0.3)						1073.1
U3 - Beta(1, 1)						1069.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.2	53.7	29.8	11.7	1.4	2.2
U2 - Beta(0.5, 0.5)	3.9	33.8	27.8	24.9	7.5	2.1
U2 - Beta(0.7, 0.3)	4.5	38.5	28	21.2	5.6	2.2
U2 - Beta(1, 1)	5	34.2	28.6	23	7	2.2
U3 - Beta(0.3, 0.7)	2.6	58.7	26.4	9.6	0.5	2.2
U3 - Beta(0.5, 0.5)	6.1	42.4	26.8	17.8	4.7	2.2
U3 - Beta(0.7, 0.3)	5.2	44.3	25	19.5	3.8	2.2
U3 - Beta(1, 1)	6.5	45.6	24.5	17.5	3.7	2.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.8	15.6	6.8	2.2	0.3	-
U2 - Beta(0.5, 0.5)	4.2	8.5	7.7	6.6	2.7	-
U2 - Beta(0.7, 0.3)	4.2	7.3	7.4	7.2	3.5	-
U2 - Beta(1, 1)	4.2	8.5	7.7	6.7	2.7	-
U3 - Beta(0.3, 0.7)	5	16.2	6.3	2	0.2	-
U3 - Beta(0.5, 0.5)	4.3	9.4	7.7	6.1	2.2	-
U3 - Beta(0.7, 0.3)	4.3	8.4	7.4	6.7	2.9	-
U3 - Beta(1, 1)	4.4	9.8	7.6	5.9	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	6.8	2.5	0.9	0.1	-
U2 - Beta(0.5, 0.5)	0.4	4.3	2.6	2.5	1	-
U2 - Beta(0.7, 0.3)	0.4	4	2.3	2.7	1.3	-
U2 - Beta(1, 1)	0.4	4.3	2.6	2.5	1	-
U3 - Beta(0.3, 0.7)	0.8	7.1	2.4	0.8	0.1	-
U3 - Beta(0.5, 0.5)	0.5	4.6	2.6	2.3	0.8	-
U3 - Beta(0.7, 0.3)	0.4	4.3	2.4	2.5	1.1	-
U3 - Beta(1, 1)	0.5	4.8	2.6	2.3	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1024
U2 - Beta(0.5, 0.5)						1038.5
U2 - Beta(0.7, 0.3)						1043
U2 - Beta(1, 1)						1039.3
U3 - Beta(0.3, 0.7)						1023
U3 - Beta(0.5, 0.5)						1036.2
U3 - Beta(0.7, 0.3)						1039.9
U3 - Beta(1, 1)						1035.2

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	66.5	5.6	0.5	0	0	27.4
U2 - Beta(0.5, 0.5)	63.4	9.2	1.1	0.2	0	26.1
U2 - Beta(0.7, 0.3)	63.3	9.8	0.8	0	0	26.1
U2 - Beta(1, 1)	63.5	8.8	1.2	0.3	0	26.2
U3 - Beta(0.3, 0.7)	66.8	5.5	0.1	0	0	27.6
U3 - Beta(0.5, 0.5)	64.7	8.2	0.6	0.2	0	26.3
U3 - Beta(0.7, 0.3)	65.4	8.1	0.7	0.1	0	25.7
U3 - Beta(1, 1)	64.1	8.6	0.8	0.1	0	26.4
Mean administrations						
U2 - Beta(0.3, 0.7)	22.8	2.6	0.2	0	0	-
U2 - Beta(0.5, 0.5)	19.7	5.1	1	0.2	0	-
U2 - Beta(0.7, 0.3)	19.1	5.3	1.1	0.4	0.1	-
U2 - Beta(1, 1)	19.7	5	1	0.2	0	-
U3 - Beta(0.3, 0.7)	22.9	2.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	20.1	4.9	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	19.5	5	1.1	0.3	0	-
U3 - Beta(1, 1)	20.1	4.9	0.8	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.5	1.2	0.1	0	0	-
U2 - Beta(0.5, 0.5)	5.2	2.2	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	5.1	2.2	0.3	0.2	0	-
U2 - Beta(1, 1)	5.2	2.2	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	6.5	1.2	0	0	0	-
U3 - Beta(0.5, 0.5)	5.3	2.1	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	5.1	2.1	0.4	0.1	0	-
U3 - Beta(1, 1)	5.3	2.1	0.3	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						982.7
U2 - Beta(0.5, 0.5)						1003.1
U2 - Beta(0.7, 0.3)						1007
U2 - Beta(1, 1)						1002.2
U3 - Beta(0.3, 0.7)						979.9
U3 - Beta(0.5, 0.5)						1003.6
U3 - Beta(0.7, 0.3)						1005.9
U3 - Beta(1, 1)						1000.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.8	64.4	2.9	0.1	0	3.8
U2 - Beta(0.5, 0.5)	18.4	61.7	14.4	2.3	0.4	2.8
U2 - Beta(0.7, 0.3)	19.4	52.1	18.7	5.7	1.3	2.8
U2 - Beta(1, 1)	18.1	62.3	14.5	2.2	0.4	2.5
U3 - Beta(0.3, 0.7)	32	62.2	1.9	0.1	0	3.8
U3 - Beta(0.5, 0.5)	19.1	64.5	12.2	1.5	0.1	2.6
U3 - Beta(0.7, 0.3)	19.7	53.7	18.4	5	0.5	2.7
U3 - Beta(1, 1)	18.8	65.2	11.6	1.5	0.1	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	14.4	14.3	0.7	0	0	-
U2 - Beta(0.5, 0.5)	9.8	15.4	3.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	9.5	11.8	5.3	2.4	0.6	-
U2 - Beta(1, 1)	9.7	15.5	3.6	0.7	0.1	-
U3 - Beta(0.3, 0.7)	14.8	14.2	0.4	0	0	-
U3 - Beta(0.5, 0.5)	10.2	15.8	3.1	0.5	0	-
U3 - Beta(0.7, 0.3)	9.7	13.2	4.7	1.7	0.2	-
U3 - Beta(1, 1)	10.2	15.8	3	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	5.1	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.2	5.6	1	0.3	0	-
U2 - Beta(0.7, 0.3)	2.2	4.4	1.4	0.8	0.2	-
U2 - Beta(1, 1)	2.2	5.6	1.1	0.2	0	-
U3 - Beta(0.3, 0.7)	4.4	5.1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.4	5.7	0.9	0.2	0	-
U3 - Beta(0.7, 0.3)	2.2	4.8	1.4	0.6	0.1	-
U3 - Beta(1, 1)	2.4	5.7	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1095.2
U2 - Beta(0.5, 0.5)						1131.3
U2 - Beta(0.7, 0.3)						1138.5
U2 - Beta(1, 1)						1132
U3 - Beta(0.3, 0.7)						1093.4
U3 - Beta(0.5, 0.5)						1128.8
U3 - Beta(0.7, 0.3)						1135
U3 - Beta(1, 1)						1128.9
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26	2.7	0.4	0	0	70.9
U2 - Beta(0.5, 0.5)	27.2	3.2	0.3	0	0	69.3
U2 - Beta(0.7, 0.3)	27.7	2.5	0.6	0	0	69.2
U2 - Beta(1, 1)	27	3.2	0.3	0	0	69.5
U3 - Beta(0.3, 0.7)	26	2.7	0.3	0	0	71
U3 - Beta(0.5, 0.5)	27.6	2.6	0.2	0	0	69.6
U3 - Beta(0.7, 0.3)	28	2.6	0.1	0	0	69.3
U3 - Beta(1, 1)	27.3	2.7	0.1	0	0	69.9
Mean administrations						
U2 - Beta(0.3, 0.7)	15	3.7	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	14.4	4.3	0.8	0.2	0	-
U2 - Beta(0.7, 0.3)	14.4	4.3	0.8	0.3	0	-
U2 - Beta(1, 1)	14.4	4.3	0.8	0.2	0	-
U3 - Beta(0.3, 0.7)	15.1	3.7	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	14.5	4.3	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	14.4	4.3	0.8	0.3	0	-
U3 - Beta(1, 1)	14.5	4.3	0.7	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.4	2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	4.1	2.3	0.3	0.1	0	-
U2 - Beta(0.7, 0.3)	4.1	2.2	0.3	0.1	0	-
U2 - Beta(1, 1)	4.1	2.3	0.3	0.1	0	-
U3 - Beta(0.3, 0.7)	4.4	2	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.1	2.2	0.3	0.1	0	-
U3 - Beta(0.7, 0.3)	4.1	2.2	0.3	0.1	0	-
U3 - Beta(1, 1)	4.1	2.3	0.2	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						693.7
U2 - Beta(0.5, 0.5)						712.9
U2 - Beta(0.7, 0.3)						714.6
U2 - Beta(1, 1)						714.1
U3 - Beta(0.3, 0.7)						691.4
U3 - Beta(0.5, 0.5)						709.6
U3 - Beta(0.7, 0.3)						710.3
U3 - Beta(1, 1)						711.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.6	15.1	27.3	14.7	5.4	24.9
U2 - Beta(0.5, 0.5)	17.4	15.4	21.7	11.9	10.6	23
U2 - Beta(0.7, 0.3)	16.2	16.1	21.6	12.2	10.7	23.2
U2 - Beta(1, 1)	18.2	15.2	20.4	12	10.8	23.4
U3 - Beta(0.3, 0.7)	16.5	16.8	24.2	13.4	3.7	25.4
U3 - Beta(0.5, 0.5)	19.9	16.9	19.5	11.2	9.2	23.3
U3 - Beta(0.7, 0.3)	20	18	18.5	10.8	9.2	23.5
U3 - Beta(1, 1)	21.5	18.1	16.9	10.5	9.3	23.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	7	7.1	3.6	0.8	-
U2 - Beta(0.5, 0.5)	8.1	6.6	5.9	4.2	2.3	-
U2 - Beta(0.7, 0.3)	8.1	6.6	5.6	4.2	2.6	-
U2 - Beta(1, 1)	8	6.7	5.8	4.2	2.3	-
U3 - Beta(0.3, 0.7)	8.9	7.3	6.7	3.3	0.6	-
U3 - Beta(0.5, 0.5)	8.4	6.6	5.8	4.2	2.1	-
U3 - Beta(0.7, 0.3)	8.4	6.8	5.4	4.2	2.2	-
U3 - Beta(1, 1)	8.5	6.8	5.5	4.1	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.8	2.7	1.5	0.3	-
U2 - Beta(0.5, 0.5)	2	3.7	2.1	1.7	0.8	-
U2 - Beta(0.7, 0.3)	2	3.7	1.9	1.7	0.8	-
U2 - Beta(1, 1)	1.9	3.8	2	1.7	0.8	-
U3 - Beta(0.3, 0.7)	2.4	3.8	2.5	1.4	0.2	-
U3 - Beta(0.5, 0.5)	2.1	3.7	2	1.7	0.7	-
U3 - Beta(0.7, 0.3)	2.1	3.8	1.9	1.6	0.8	-
U3 - Beta(1, 1)	2.2	3.9	1.9	1.6	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						907.4
U2 - Beta(0.5, 0.5)						929.8
U2 - Beta(0.7, 0.3)						932.2
U2 - Beta(1, 1)						930.9
U3 - Beta(0.3, 0.7)						904
U3 - Beta(0.5, 0.5)						927.8
U3 - Beta(0.7, 0.3)						928.6
U3 - Beta(1, 1)						924.8

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	26.2	3.4	0	0	56.6
U2 - Beta(0.5, 0.5)	14.8	23.1	7.6	1.2	0.1	53.2
U2 - Beta(0.7, 0.3)	13.9	23.8	7.3	1.6	0.3	53.1
U2 - Beta(1, 1)	14.2	23.4	7.2	1.2	0.1	53.9
U3 - Beta(0.3, 0.7)	13.7	26	3.1	0	0	57.2
U3 - Beta(0.5, 0.5)	14	24.2	6.9	1.1	0.1	53.7
U3 - Beta(0.7, 0.3)	14.4	24.3	7.1	1.7	0	52.5
U3 - Beta(1, 1)	14.2	24.2	6.6	1.2	0	53.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	9.7	1.1	0	0	-
U2 - Beta(0.5, 0.5)	10.7	8.6	2.6	0.7	0	-
U2 - Beta(0.7, 0.3)	10.5	8	2.9	1.1	0.2	-
U2 - Beta(1, 1)	10.6	8.7	2.6	0.7	0	-
U3 - Beta(0.3, 0.7)	11.2	9.7	1.1	0	0	-
U3 - Beta(0.5, 0.5)	10.8	8.8	2.4	0.6	0	-
U3 - Beta(0.7, 0.3)	10.7	8.1	2.7	1	0.1	-
U3 - Beta(1, 1)	10.8	8.8	2.4	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	4.3	0.4	0	0	-
U2 - Beta(0.5, 0.5)	2.7	3.9	0.8	0.3	0	-
U2 - Beta(0.7, 0.3)	2.6	3.7	0.8	0.4	0.1	-
U2 - Beta(1, 1)	2.6	3.9	0.8	0.3	0	-
U3 - Beta(0.3, 0.7)	2.9	4.3	0.4	0	0	-
U3 - Beta(0.5, 0.5)	2.7	4	0.7	0.2	0	-
U3 - Beta(0.7, 0.3)	2.6	3.8	0.8	0.4	0	-
U3 - Beta(1, 1)	2.7	4	0.7	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						793.6
U2 - Beta(0.5, 0.5)						819.9
U2 - Beta(0.7, 0.3)						827.3
U2 - Beta(1, 1)						820.1
U3 - Beta(0.3, 0.7)						790.6
U3 - Beta(0.5, 0.5)						819.9
U3 - Beta(0.7, 0.3)						825
U3 - Beta(1, 1)						819.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	70.1	14.1	0.6	0	0	15.2
U2 - Beta(0.5, 0.5)	69.5	14.4	0.8	0.1	0	15.2
U2 - Beta(0.7, 0.3)	68.4	15.4	0.8	0.1	0	15.3
U2 - Beta(1, 1)	68.6	15.3	0.6	0.1	0	15.4
U3 - Beta(0.3, 0.7)	70.8	13.7	0	0	0	15.5
U3 - Beta(0.5, 0.5)	71.9	12.6	0.4	0	0	15.1
U3 - Beta(0.7, 0.3)	70.1	14.6	0.5	0.1	0	14.7
U3 - Beta(1, 1)	70.2	14.3	0.5	0	0	15
Mean administrations						
U2 - Beta(0.3, 0.7)	20.8	5.9	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	18.9	6.8	1.7	0.5	0	-
U2 - Beta(0.7, 0.3)	18.6	7	1.7	0.6	0	-
U2 - Beta(1, 1)	18.5	7.1	1.7	0.5	0	-
U3 - Beta(0.3, 0.7)	21.3	5.7	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	19.2	6.6	1.5	0.4	0	-
U3 - Beta(0.7, 0.3)	18.9	6.8	1.6	0.5	0	-
U3 - Beta(1, 1)	19.2	6.7	1.5	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.5	2.8	0.4	0	0	-
U2 - Beta(0.5, 0.5)	5.6	3.2	0.6	0.2	0	-
U2 - Beta(0.7, 0.3)	5.5	3.3	0.5	0.2	0	-
U2 - Beta(1, 1)	5.5	3.4	0.6	0.2	0	-
U3 - Beta(0.3, 0.7)	6.7	2.8	0.2	0	0	-
U3 - Beta(0.5, 0.5)	5.7	3.2	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	5.7	3.2	0.5	0.2	0	-
U3 - Beta(1, 1)	5.8	3.2	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						988.2
U2 - Beta(0.5, 0.5)						1000.7
U2 - Beta(0.7, 0.3)						1003.1
U2 - Beta(1, 1)						999.9
U3 - Beta(0.3, 0.7)						987
U3 - Beta(0.5, 0.5)						999.4
U3 - Beta(0.7, 0.3)						1004
U3 - Beta(1, 1)						1003.1
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.8	15.7	17.9	50.4	2.6	2.6
U2 - Beta(0.5, 0.5)	11.9	16.8	16.3	42.7	9.7	2.6
U2 - Beta(0.7, 0.3)	14.6	17	15.1	41.2	9.6	2.5
U2 - Beta(1, 1)	13.5	14.7	15.3	44.6	9.3	2.6
U3 - Beta(0.3, 0.7)	14.3	18.1	13.4	49.4	1.9	2.9
U3 - Beta(0.5, 0.5)	16.6	17.7	13.4	41.8	8	2.5
U3 - Beta(0.7, 0.3)	17.8	18.7	12.4	38.9	9.6	2.6
U3 - Beta(1, 1)	19.5	15.6	12.1	42.1	8.1	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	6.9	7.4	9.5	0.8	-
U2 - Beta(0.5, 0.5)	5	6.3	6.5	9	2.9	-
U2 - Beta(0.7, 0.3)	5.1	6.2	6.4	8.2	3.8	-
U2 - Beta(1, 1)	5.1	6.4	6.4	9	2.9	-
U3 - Beta(0.3, 0.7)	5.6	7.5	7.1	8.9	0.7	-
U3 - Beta(0.5, 0.5)	5.4	6.4	6.3	8.9	2.7	-
U3 - Beta(0.7, 0.3)	5.6	6.6	6.1	8.2	3.3	-
U3 - Beta(1, 1)	5.7	6.5	6.1	8.7	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.8	2.7	3.4	0.3	-
U2 - Beta(0.5, 0.5)	0.8	3.8	2	3.2	0.9	-
U2 - Beta(0.7, 0.3)	0.9	3.8	1.9	3	1	-
U2 - Beta(1, 1)	0.9	3.8	1.9	3.2	0.9	-
U3 - Beta(0.3, 0.7)	1.2	4	2.6	3.2	0.2	-
U3 - Beta(0.5, 0.5)	1	3.9	1.9	3.2	0.8	-
U3 - Beta(0.7, 0.3)	1.1	3.9	1.8	2.9	1	-
U3 - Beta(1, 1)	1.2	3.9	1.8	3.1	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022
U2 - Beta(0.5, 0.5)						1046.4
U2 - Beta(0.7, 0.3)						1051.3
U2 - Beta(1, 1)						1044
U3 - Beta(0.3, 0.7)						1013.5
U3 - Beta(0.5, 0.5)						1041
U3 - Beta(0.7, 0.3)						1043.3
U3 - Beta(1, 1)						1038.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	25.2	6.3	1.1	0	53.6
U2 - Beta(0.5, 0.5)	14.9	23.5	8.2	1.5	0.2	51.7
U2 - Beta(0.7, 0.3)	14.2	25.9	6.8	2	0.2	50.9
U2 - Beta(1, 1)	14.5	25.4	6.8	1.7	0.1	51.5
U3 - Beta(0.3, 0.7)	13.7	26	5.6	0.8	0	53.9
U3 - Beta(0.5, 0.5)	15.1	25.5	5.8	1.5	0.1	52
U3 - Beta(0.7, 0.3)	14.7	26.3	5.8	1.9	0	51.3
U3 - Beta(1, 1)	15.1	26.4	4.9	1.6	0.1	51.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	9.5	1.9	0.3	0	-
U2 - Beta(0.5, 0.5)	10.7	8.1	3	1.2	0.2	-
U2 - Beta(0.7, 0.3)	10.4	8	3	1.5	0.3	-
U2 - Beta(1, 1)	10.6	8.2	3	1.2	0.1	-
U3 - Beta(0.3, 0.7)	11.2	9.5	1.7	0.3	0	-
U3 - Beta(0.5, 0.5)	10.7	8.3	2.9	1.1	0.1	-
U3 - Beta(0.7, 0.3)	10.7	8	2.9	1.4	0.2	-
U3 - Beta(1, 1)	10.7	8.5	2.8	1.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	4.4	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	3.9	1	0.5	0.1	-
U2 - Beta(0.7, 0.3)	2.7	3.9	0.9	0.6	0.1	-
U2 - Beta(1, 1)	2.7	4	1	0.5	0.1	-
U3 - Beta(0.3, 0.7)	3	4.4	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	2.8	4	0.9	0.5	0.1	-
U3 - Beta(0.7, 0.3)	2.7	3.9	0.9	0.5	0.1	-
U3 - Beta(1, 1)	2.8	4.1	0.9	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						798
U2 - Beta(0.5, 0.5)						818.3
U2 - Beta(0.7, 0.3)						824.7
U2 - Beta(1, 1)						819.7
U3 - Beta(0.3, 0.7)						795.7
U3 - Beta(0.5, 0.5)						818.1
U3 - Beta(0.7, 0.3)						819.5
U3 - Beta(1, 1)						818.6
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.6	20.8	22.6	29.3	3.5	9.2
U2 - Beta(0.5, 0.5)	20.2	21.2	20.5	23.6	5.6	8.9
U2 - Beta(0.7, 0.3)	20.4	19.9	21.1	23.4	5.8	9.4
U2 - Beta(1, 1)	20.3	20.8	19.5	24.2	5.9	9.3
U3 - Beta(0.3, 0.7)	19.8	23.9	16	28	2.7	9.6
U3 - Beta(0.5, 0.5)	26.1	22.8	16.5	21.1	4.5	9
U3 - Beta(0.7, 0.3)	25.4	24.3	16.5	19.8	4.7	9.3
U3 - Beta(1, 1)	25.9	23	16.3	20.9	4.2	9.7
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	7.7	7.3	6.3	0.9	-
U2 - Beta(0.5, 0.5)	6.7	7.2	6.6	6.3	2.2	-
U2 - Beta(0.7, 0.3)	6.8	7.2	6.5	6.1	2.5	-
U2 - Beta(1, 1)	6.8	7.3	6.4	6.3	2.3	-
U3 - Beta(0.3, 0.7)	7.4	8	6.7	6	0.8	-
U3 - Beta(0.5, 0.5)	7.1	7.3	6.4	6.2	2.1	-
U3 - Beta(0.7, 0.3)	7.2	7.4	6.2	6	2.3	-
U3 - Beta(1, 1)	7.4	7.5	6.1	6.1	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	4.1	2.7	2.4	0.4	-
U2 - Beta(0.5, 0.5)	1.5	4.1	2.2	2.5	0.7	-
U2 - Beta(0.7, 0.3)	1.5	4.1	2.1	2.5	0.8	-
U2 - Beta(1, 1)	1.6	4.1	2.1	2.5	0.8	-
U3 - Beta(0.3, 0.7)	2	4.2	2.5	2.3	0.3	-
U3 - Beta(0.5, 0.5)	1.7	4.2	2.1	2.5	0.7	-
U3 - Beta(0.7, 0.3)	1.8	4.2	2	2.4	0.8	-
U3 - Beta(1, 1)	1.9	4.3	1.9	2.4	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						973.5
U2 - Beta(0.5, 0.5)						987.2
U2 - Beta(0.7, 0.3)						987.6
U2 - Beta(1, 1)						986.6
U3 - Beta(0.3, 0.7)						968.7
U3 - Beta(0.5, 0.5)						984.6
U3 - Beta(0.7, 0.3)						985.3
U3 - Beta(1, 1)						982.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	39	31.9	11.4	2.1	0.1	15.5
U2 - Beta(0.5, 0.5)	43	30.5	10	1.8	0.1	14.6
U2 - Beta(0.7, 0.3)	41.1	31.9	10	1.8	0	15.2
U2 - Beta(1, 1)	40.9	31.4	10.8	1.8	0	15.1
U3 - Beta(0.3, 0.7)	44.5	32	6.6	1.5	0	15.4
U3 - Beta(0.5, 0.5)	45.3	30.3	7.6	1.1	0.1	15.6
U3 - Beta(0.7, 0.3)	45	31.2	7.7	1.2	0.1	14.8
U3 - Beta(1, 1)	44.3	31.3	6.9	1.6	0.1	15.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.4	9.5	4.9	2.1	0.4	-
U2 - Beta(0.5, 0.5)	11.2	9	4.9	2.7	0.7	-
U2 - Beta(0.7, 0.3)	11.4	8.9	4.8	2.7	0.6	-
U2 - Beta(1, 1)	11.1	9.1	4.9	2.7	0.6	-
U3 - Beta(0.3, 0.7)	12.7	9.5	4	1.8	0.3	-
U3 - Beta(0.5, 0.5)	11.8	8.8	4.6	2.6	0.6	-
U3 - Beta(0.7, 0.3)	11.8	8.9	4.6	2.5	0.6	-
U3 - Beta(1, 1)	12	8.8	4.5	2.5	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	4.6	1.9	0.8	0.2	-
U2 - Beta(0.5, 0.5)	3.3	4.6	1.7	1.2	0.3	-
U2 - Beta(0.7, 0.3)	3.4	4.6	1.7	1.2	0.2	-
U2 - Beta(1, 1)	3.3	4.6	1.7	1.2	0.3	-
U3 - Beta(0.3, 0.7)	4.1	4.6	1.6	0.7	0.1	-
U3 - Beta(0.5, 0.5)	3.6	4.5	1.6	1.1	0.3	-
U3 - Beta(0.7, 0.3)	3.6	4.5	1.6	1.1	0.2	-
U3 - Beta(1, 1)	3.7	4.5	1.5	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						943
U2 - Beta(0.5, 0.5)						955.3
U2 - Beta(0.7, 0.3)						952
U2 - Beta(1, 1)						954.5
U3 - Beta(0.3, 0.7)						940.2
U3 - Beta(0.5, 0.5)						950.8
U3 - Beta(0.7, 0.3)						953.6
U3 - Beta(1, 1)						950.8
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.5	28.2	23.5	16.6	9.2	0
U2 - Beta(0.5, 0.5)	32	30.2	15.9	14.8	7.1	0
U2 - Beta(0.7, 0.3)	31.6	33.9	15.1	12.8	6.6	0
U2 - Beta(1, 1)	33.6	30.4	16.1	12.6	7.3	0
U3 - Beta(0.3, 0.7)	28.9	32.3	20.7	12.6	5.5	0
U3 - Beta(0.5, 0.5)	37.3	33.1	13.4	11.4	4.8	0
U3 - Beta(0.7, 0.3)	34	37.7	14.2	10.1	4	0
U3 - Beta(1, 1)	40	34	13.9	7.9	4.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	10.1	9	5.8	3.5	1.6	-
U2 - Beta(0.5, 0.5)	4.5	5.7	6.2	8.1	5.6	-
U2 - Beta(0.7, 0.3)	4.3	5.2	6	8.5	6	-
U2 - Beta(1, 1)	4.5	5.8	6.3	7.9	5.5	-
U3 - Beta(0.3, 0.7)	10.7	9.8	5.2	3	1.3	-
U3 - Beta(0.5, 0.5)	4.8	6.2	6.4	7.7	4.9	-
U3 - Beta(0.7, 0.3)	5	6	6.1	7.9	5	-
U3 - Beta(1, 1)	5.1	6.5	6.5	7.3	4.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	4.2	2.5	1.5	0.7	-
U2 - Beta(0.5, 0.5)	0.6	3.1	2.4	3.4	2.2	-
U2 - Beta(0.7, 0.3)	0.5	3	2.2	3.7	2.2	-
U2 - Beta(1, 1)	0.6	3.2	2.5	3.3	2.1	-
U3 - Beta(0.3, 0.7)	3.4	4.6	2.2	1.3	0.5	-
U3 - Beta(0.5, 0.5)	0.8	3.3	2.5	3.3	1.9	-
U3 - Beta(0.7, 0.3)	0.8	3.3	2.3	3.4	1.9	-
U3 - Beta(1, 1)	0.9	3.5	2.5	3.1	1.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						987.4
U2 - Beta(0.5, 0.5)						1007.1
U2 - Beta(0.7, 0.3)						1009.5
U2 - Beta(1, 1)						1006.5
U3 - Beta(0.3, 0.7)						983.7
U3 - Beta(0.5, 0.5)						1004.4
U3 - Beta(0.7, 0.3)						1005.4
U3 - Beta(1, 1)						1003.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	56.7	42	0.8	0	0	0.5
U2 - Beta(0.5, 0.5)	22.8	69.3	7.2	0.1	0	0.6
U2 - Beta(0.7, 0.3)	24.7	63.2	11.5	0.2	0	0.4
U2 - Beta(1, 1)	22	69.8	7.7	0.1	0	0.4
U3 - Beta(0.3, 0.7)	59.1	39.9	0.5	0	0	0.5
U3 - Beta(0.5, 0.5)	25.4	69.3	4.9	0.1	0	0.3
U3 - Beta(0.7, 0.3)	24.9	64.7	9.4	0.5	0	0.5
U3 - Beta(1, 1)	26	68.8	4.8	0.1	0	0.3
Mean administrations						
U2 - Beta(0.3, 0.7)	20.2	9.4	0.3	0	0	-
U2 - Beta(0.5, 0.5)	10.9	16.6	2.3	0.2	0	-
U2 - Beta(0.7, 0.3)	10.3	14.1	4.5	1.1	0.1	-
U2 - Beta(1, 1)	10.8	16.6	2.4	0.2	0	-
U3 - Beta(0.3, 0.7)	20.5	9.3	0.2	0	0	-
U3 - Beta(0.5, 0.5)	11.5	16.6	1.7	0.1	0	-
U3 - Beta(0.7, 0.3)	10.4	15.3	3.7	0.6	0	-
U3 - Beta(1, 1)	11.6	16.6	1.7	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	3.3	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.5	5.6	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	2.3	4.7	1.2	0.4	0	-
U2 - Beta(1, 1)	2.5	5.6	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	6.5	3.3	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.8	5.6	0.5	0	0	-
U3 - Beta(0.7, 0.3)	2.3	5.1	1.1	0.2	0	-
U3 - Beta(1, 1)	2.8	5.6	0.5	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.4
U2 - Beta(0.5, 0.5)						1162.8
U2 - Beta(0.7, 0.3)						1178.4
U2 - Beta(1, 1)						1166.3
U3 - Beta(0.3, 0.7)						1107.7
U3 - Beta(0.5, 0.5)						1161.1
U3 - Beta(0.7, 0.3)						1171.5
U3 - Beta(1, 1)						1161
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.6	9.3	15.2	33.3	38.4	0.2
U2 - Beta(0.5, 0.5)	9.9	11.1	13.1	21.2	44.5	0.2
U2 - Beta(0.7, 0.3)	10.2	15.1	12.9	19	42.6	0.2
U2 - Beta(1, 1)	11.4	11.7	12.2	18.9	45.6	0.2
U3 - Beta(0.3, 0.7)	8.7	11.7	18.3	32.2	29	0.1
U3 - Beta(0.5, 0.5)	13.6	15.1	13	17	41.2	0.1
U3 - Beta(0.7, 0.3)	16.4	16.7	12.3	15.8	38.6	0.2
U3 - Beta(1, 1)	18.3	16.4	11.6	14.4	39.1	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	3.9	5.6	7	8.6	5	-
U2 - Beta(0.5, 0.5)	3.5	4.2	5.5	7.8	8.9	-
U2 - Beta(0.7, 0.3)	3.5	4.2	5.4	7.6	9.2	-
U2 - Beta(1, 1)	3.5	4.4	5.5	7.7	9	-
U3 - Beta(0.3, 0.7)	4	5.8	7.3	8.6	4.3	-
U3 - Beta(0.5, 0.5)	3.6	4.4	5.6	7.9	8.5	-
U3 - Beta(0.7, 0.3)	3.7	4.5	5.5	7.6	8.7	-
U3 - Beta(1, 1)	3.7	4.6	5.6	7.7	8.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	3.3	2.8	3.4	1.7	-
U2 - Beta(0.5, 0.5)	0.2	2.9	1.9	3.1	2.7	-
U2 - Beta(0.7, 0.3)	0.2	2.9	1.8	3.1	2.7	-
U2 - Beta(1, 1)	0.2	3	1.9	3.1	2.7	-
U3 - Beta(0.3, 0.7)	0.5	3.4	2.9	3.4	1.5	-
U3 - Beta(0.5, 0.5)	0.3	3	1.9	3.2	2.6	-
U3 - Beta(0.7, 0.3)	0.3	3.1	1.9	3	2.6	-
U3 - Beta(1, 1)	0.3	3.1	1.9	3.1	2.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1006.3
U2 - Beta(0.5, 0.5)						1050.6
U2 - Beta(0.7, 0.3)						1052.9
U2 - Beta(1, 1)						1050.6
U3 - Beta(0.3, 0.7)						999.5
U3 - Beta(0.5, 0.5)						1046.7
U3 - Beta(0.7, 0.3)						1049.4
U3 - Beta(1, 1)						1045

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.1	58.1	8.8	0.1	0	16.9
U2 - Beta(0.5, 0.5)	16.6	56.4	9.6	0.2	0	17.2
U2 - Beta(0.7, 0.3)	18.2	55.2	8.6	0.6	0	17.4
U2 - Beta(1, 1)	16	56.3	10	0.4	0	17.3
U3 - Beta(0.3, 0.7)	18.7	58.3	6.2	0.1	0	16.7
U3 - Beta(0.5, 0.5)	19	54.5	7.9	0.3	0	18.3
U3 - Beta(0.7, 0.3)	18.6	56.4	7.3	0.5	0	17.2
U3 - Beta(1, 1)	18.3	55.2	8.6	0.3	0	17.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9.7	15.4	2.8	0.3	0	-
U2 - Beta(0.5, 0.5)	9.5	13.1	4.6	1	0.1	-
U2 - Beta(0.7, 0.3)	9.4	12.4	4.8	1.6	0.1	-
U2 - Beta(1, 1)	9.3	13.2	4.7	1	0.1	-
U3 - Beta(0.3, 0.7)	10.3	15.4	2.3	0.2	0	-
U3 - Beta(0.5, 0.5)	9.7	13.3	4.2	1	0	-
U3 - Beta(0.7, 0.3)	9.6	12.7	4.5	1.3	0.1	-
U3 - Beta(1, 1)	9.7	13.3	4.3	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	6.5	1	0.1	0	-
U2 - Beta(0.5, 0.5)	2.3	5.7	1.4	0.5	0	-
U2 - Beta(0.7, 0.3)	2.2	5.6	1.4	0.6	0	-
U2 - Beta(1, 1)	2.2	5.8	1.5	0.5	0	-
U3 - Beta(0.3, 0.7)	2.8	6.5	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	2.4	5.9	1.3	0.4	0	-
U3 - Beta(0.7, 0.3)	2.3	5.7	1.3	0.5	0	-
U3 - Beta(1, 1)	2.4	5.8	1.3	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						989.2
U2 - Beta(0.5, 0.5)						1004.2
U2 - Beta(0.7, 0.3)						1010.2
U2 - Beta(1, 1)						1004.8
U3 - Beta(0.3, 0.7)						984.8
U3 - Beta(0.5, 0.5)						999.9
U3 - Beta(0.7, 0.3)						1007.9
U3 - Beta(1, 1)						1002
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.1	57.5	29.2	3.7	0.1	3.4
U2 - Beta(0.5, 0.5)	8.2	56.3	27.1	5.4	0	3
U2 - Beta(0.7, 0.3)	9.4	54.7	26.4	6.4	0.1	3
U2 - Beta(1, 1)	8.9	54.8	27.1	6.4	0	2.8
U3 - Beta(0.3, 0.7)	8.3	62.7	22.3	3.3	0.1	3.3
U3 - Beta(0.5, 0.5)	11.7	58.6	22.2	4.7	0.1	2.7
U3 - Beta(0.7, 0.3)	11.3	57.9	22	5.4	0.2	3.2
U3 - Beta(1, 1)	12.5	58.7	20	5.9	0	2.9
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	15	7.1	1.8	0.2	-
U2 - Beta(0.5, 0.5)	5.3	11.9	8.1	3.6	0.7	-
U2 - Beta(0.7, 0.3)	5.4	11.6	7.8	3.9	0.9	-
U2 - Beta(1, 1)	5.3	11.9	8.1	3.6	0.7	-
U3 - Beta(0.3, 0.7)	5.8	16	6.4	1.3	0.1	-
U3 - Beta(0.5, 0.5)	5.6	12.4	7.7	3.2	0.6	-
U3 - Beta(0.7, 0.3)	5.6	12.1	7.4	3.7	0.8	-
U3 - Beta(1, 1)	5.8	12.5	7.4	3.3	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	6.6	2.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	0.9	5.5	2.6	1.4	0.3	-
U2 - Beta(0.7, 0.3)	0.9	5.5	2.5	1.5	0.3	-
U2 - Beta(1, 1)	0.9	5.5	2.7	1.4	0.3	-
U3 - Beta(0.3, 0.7)	1.2	7	2.3	0.6	0.1	-
U3 - Beta(0.5, 0.5)	1	5.8	2.5	1.3	0.2	-
U3 - Beta(0.7, 0.3)	1	5.6	2.4	1.4	0.3	-
U3 - Beta(1, 1)	1.1	5.8	2.4	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1018.9
U2 - Beta(0.5, 0.5)						1035.9
U2 - Beta(0.7, 0.3)						1038.3
U2 - Beta(1, 1)						1036.6
U3 - Beta(0.3, 0.7)						1016.8
U3 - Beta(0.5, 0.5)						1032.6
U3 - Beta(0.7, 0.3)						1034.9
U3 - Beta(1, 1)						1032.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	41	0	0	0	0	59
U2 - Beta(0.5, 0.5)	37.1	0.1	0	0	0	62.8
U2 - Beta(0.7, 0.3)	34.8	0.2	0	0	0	65
U2 - Beta(1, 1)	37.4	0	0	0	0	62.6
U3 - Beta(0.3, 0.7)	40.7	0	0	0	0	59.3
U3 - Beta(0.5, 0.5)	37	0.1	0	0	0	62.9
U3 - Beta(0.7, 0.3)	36.1	0.1	0	0	0	63.8
U3 - Beta(1, 1)	37.9	0	0	0	0	62.1
Mean administrations						
U2 - Beta(0.3, 0.7)	20.2	0.3	0	0	0	-
U2 - Beta(0.5, 0.5)	18.2	1.7	0	0	0	-
U2 - Beta(0.7, 0.3)	16.8	2.9	0	0	0	-
U2 - Beta(1, 1)	18.2	1.7	0	0	0	-
U3 - Beta(0.3, 0.7)	20.2	0.3	0	0	0	-
U3 - Beta(0.5, 0.5)	18.3	1.6	0	0	0	-
U3 - Beta(0.7, 0.3)	17.5	2.2	0	0	0	-
U3 - Beta(1, 1)	18.3	1.6	0	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.3	0.2	0	0	0	-
U2 - Beta(0.5, 0.5)	3.5	0.8	0	0	0	-
U2 - Beta(0.7, 0.3)	3.1	1	0	0	0	-
U2 - Beta(1, 1)	3.5	0.8	0	0	0	-
U3 - Beta(0.3, 0.7)	4.3	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.5	0.8	0	0	0	-
U3 - Beta(0.7, 0.3)	3.3	1	0	0	0	-
U3 - Beta(1, 1)	3.6	0.8	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						881.9
U2 - Beta(0.5, 0.5)						858.7
U2 - Beta(0.7, 0.3)						854.4
U2 - Beta(1, 1)						858.8
U3 - Beta(0.3, 0.7)						880.3
U3 - Beta(0.5, 0.5)						859.9
U3 - Beta(0.7, 0.3)						853.9
U3 - Beta(1, 1)						860.9
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	16.8	63.1	6.4	0.4	0	13.3
U2 - Beta(0.5, 0.5)	19.8	59.5	6.9	0.2	0	13.6
U2 - Beta(0.7, 0.3)	19.1	59.4	6.8	0.4	0	14.3
U2 - Beta(1, 1)	19.5	58.1	8	0.3	0	14.1
U3 - Beta(0.3, 0.7)	21.3	60.4	4.6	0.1	0	13.6
U3 - Beta(0.5, 0.5)	19.8	59.6	6.6	0.4	0	13.6
U3 - Beta(0.7, 0.3)	21	57.2	7.1	0.4	0	14.3
U3 - Beta(1, 1)	21.2	56.2	7.4	0.5	0	14.7
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	15.9	3.1	0.3	0	-
U2 - Beta(0.5, 0.5)	9.8	13.6	4.3	1.1	0	-
U2 - Beta(0.7, 0.3)	9.8	13	4.5	1.4	0	-
U2 - Beta(1, 1)	9.7	13.5	4.4	1.2	0	-
U3 - Beta(0.3, 0.7)	10.2	15.9	2.5	0.2	0	-
U3 - Beta(0.5, 0.5)	10	13.5	4.1	1.1	0	-
U3 - Beta(0.7, 0.3)	10	13.2	4.3	1.3	0	-
U3 - Beta(1, 1)	10	13.5	4.1	1.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	6.7	1.2	0.1	0	-
U2 - Beta(0.5, 0.5)	2.5	6.1	1.3	0.5	0	-
U2 - Beta(0.7, 0.3)	2.5	6	1.2	0.6	0	-
U2 - Beta(1, 1)	2.4	6.1	1.3	0.5	0	-
U3 - Beta(0.3, 0.7)	2.9	6.8	1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.6	6.1	1.2	0.4	0	-
U3 - Beta(0.7, 0.3)	2.6	6	1.2	0.5	0	-
U3 - Beta(1, 1)	2.6	6.1	1.2	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1000.3
U2 - Beta(0.5, 0.5)						1007.9
U2 - Beta(0.7, 0.3)						1011.1
U2 - Beta(1, 1)						1007
U3 - Beta(0.3, 0.7)						992.4
U3 - Beta(0.5, 0.5)						1001.6
U3 - Beta(0.7, 0.3)						1008.2
U3 - Beta(1, 1)						1002

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.6	0	0	0	0	94.4
U2 - Beta(0.5, 0.5)	5.5	0.1	0	0	0	94.4
U2 - Beta(0.7, 0.3)	5.4	0.1	0	0	0	94.5
U2 - Beta(1, 1)	5.4	0.1	0	0	0	94.5
U3 - Beta(0.3, 0.7)	5.6	0	0	0	0	94.4
U3 - Beta(0.5, 0.5)	5.5	0.2	0	0	0	94.3
U3 - Beta(0.7, 0.3)	5.6	0.1	0	0	0	94.3
U3 - Beta(1, 1)	5.5	0.1	0	0	0	94.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	2.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	10.8	3.2	0.2	0	0	-
U2 - Beta(0.7, 0.3)	10.7	3.2	0.2	0	0	-
U2 - Beta(1, 1)	10.7	3.2	0.2	0	0	-
U3 - Beta(0.3, 0.7)	11.1	2.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	10.8	3.2	0.2	0	0	-
U3 - Beta(0.7, 0.3)	10.8	3.2	0.2	0	0	-
U3 - Beta(1, 1)	10.8	3.2	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	1.8	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.9	1.9	0.1	0	0	-
U2 - Beta(0.7, 0.3)	2.8	1.9	0.1	0	0	-
U2 - Beta(1, 1)	2.8	1.9	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3	1.8	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	1.9	0.1	0	0	-
U3 - Beta(0.7, 0.3)	2.9	1.9	0.1	0	0	-
U3 - Beta(1, 1)	2.9	1.9	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						505.1
U2 - Beta(0.5, 0.5)						504.2
U2 - Beta(0.7, 0.3)						503.1
U2 - Beta(1, 1)						502.7
U3 - Beta(0.3, 0.7)						503.3
U3 - Beta(0.5, 0.5)						504.8
U3 - Beta(0.7, 0.3)						502.9
U3 - Beta(1, 1)						503.6
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.4	1.5	15.5	73.4	9.2	0
U2 - Beta(0.5, 0.5)	0.4	0.5	1.4	63.8	33.9	0
U2 - Beta(0.7, 0.3)	0.7	2.8	4.3	53	39.2	0
U2 - Beta(1, 1)	0.4	0.6	1	64.2	33.8	0
U3 - Beta(0.3, 0.7)	0.4	1.5	18.5	74.1	5.5	0
U3 - Beta(0.5, 0.5)	1.1	1.1	2.3	67.4	28.1	0
U3 - Beta(0.7, 0.3)	1.6	3.5	4.6	53.4	36.9	0
U3 - Beta(1, 1)	1.3	1.4	2	68.1	27.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	4.5	7.9	12.7	1.2	-
U2 - Beta(0.5, 0.5)	3.1	3.2	3.8	13.9	6	-
U2 - Beta(0.7, 0.3)	3.1	3.1	3.7	11.4	8.8	-
U2 - Beta(1, 1)	3.1	3.2	3.8	14	6	-
U3 - Beta(0.3, 0.7)	3.7	4.6	8.3	12.7	0.8	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	14.5	5.3	-
U3 - Beta(0.7, 0.3)	3.2	3.2	3.8	11.8	8	-
U3 - Beta(1, 1)	3.1	3.2	4	14.5	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	3.2	4.4	0.4	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.8	1.8	-
U2 - Beta(0.7, 0.3)	0	2.4	1.1	4	2.5	-
U2 - Beta(1, 1)	0	2.4	1.2	4.8	1.8	-
U3 - Beta(0.3, 0.7)	0.4	2.8	3.4	4.4	0.3	-
U3 - Beta(0.5, 0.5)	0	2.5	1.3	5	1.6	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.2	4.1	2.4	-
U3 - Beta(1, 1)	0	2.5	1.3	5	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1033.8
U2 - Beta(0.5, 0.5)						1087.6
U2 - Beta(0.7, 0.3)						1096.2
U2 - Beta(1, 1)						1088.6
U3 - Beta(0.3, 0.7)						1030.9
U3 - Beta(0.5, 0.5)						1085.3
U3 - Beta(0.7, 0.3)						1090.4
U3 - Beta(1, 1)						1084.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	29.8	49.1	19.6	0.2	0	1.3
U2 - Beta(0.5, 0.5)	27.9	41.4	28.5	1	0	1.2
U2 - Beta(0.7, 0.3)	30.6	41.1	24.8	2.3	0	1.2
U2 - Beta(1, 1)	29.2	38.3	29.9	1.4	0	1.2
U3 - Beta(0.3, 0.7)	37.4	48.5	12.9	0.1	0	1.1
U3 - Beta(0.5, 0.5)	33.1	42	22.8	0.9	0	1.2
U3 - Beta(0.7, 0.3)	33	41.3	22.3	1.9	0	1.5
U3 - Beta(1, 1)	32.7	41.5	23.6	1	0	1.2
Mean administrations						
U2 - Beta(0.3, 0.7)	12.8	12.3	4.5	0.3	0	-
U2 - Beta(0.5, 0.5)	9.6	11.3	7.6	1.3	0.1	-
U2 - Beta(0.7, 0.3)	9.9	10.9	6.8	2.2	0.1	-
U2 - Beta(1, 1)	9.4	11.2	7.8	1.4	0.1	-
U3 - Beta(0.3, 0.7)	14.1	12.3	3.3	0.2	0	-
U3 - Beta(0.5, 0.5)	10.3	11.3	7	1.2	0	-
U3 - Beta(0.7, 0.3)	10.2	11.1	6.6	1.9	0.1	-
U3 - Beta(1, 1)	10.4	11.4	6.9	1.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4	5.1	1.5	0.1	0	-
U2 - Beta(0.5, 0.5)	2.4	4.8	2.4	0.5	0	-
U2 - Beta(0.7, 0.3)	2.6	4.7	1.9	0.7	0.1	-
U2 - Beta(1, 1)	2.4	4.8	2.4	0.6	0	-
U3 - Beta(0.3, 0.7)	4.5	5.2	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.8	4.8	2.2	0.5	0	-
U3 - Beta(0.7, 0.3)	2.7	4.7	1.9	0.7	0	-
U3 - Beta(1, 1)	2.8	4.9	2.1	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1050.1
U2 - Beta(0.5, 0.5)						1088.3
U2 - Beta(0.7, 0.3)						1094.7
U2 - Beta(1, 1)						1090.1
U3 - Beta(0.3, 0.7)						1042.1
U3 - Beta(0.5, 0.5)						1082.1
U3 - Beta(0.7, 0.3)						1086
U3 - Beta(1, 1)						1080.2

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.7	39.7	21.4	0.6	0	15.6
U2 - Beta(0.5, 0.5)	25.6	36.5	20.5	1.8	0	15.6
U2 - Beta(0.7, 0.3)	26.5	39	17.3	2	0	15.2
U2 - Beta(1, 1)	25.5	36.7	21.1	1.8	0	14.9
U3 - Beta(0.3, 0.7)	28.3	38.7	17.1	0.5	0	15.4
U3 - Beta(0.5, 0.5)	28.1	36.4	18.4	1.5	0	15.6
U3 - Beta(0.7, 0.3)	28.9	37.1	16.3	1.8	0	15.9
U3 - Beta(1, 1)	27.2	36.6	18.6	1.5	0	16.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10.6	11.4	5.4	0.5	0	-
U2 - Beta(0.5, 0.5)	10.4	9.9	6.2	1.6	0.1	-
U2 - Beta(0.7, 0.3)	10.5	9.9	5.6	2	0.2	-
U2 - Beta(1, 1)	10.2	10	6.2	1.7	0.1	-
U3 - Beta(0.3, 0.7)	11.6	11.5	4.6	0.3	0	-
U3 - Beta(0.5, 0.5)	10.7	10	5.9	1.5	0	-
U3 - Beta(0.7, 0.3)	10.7	9.9	5.5	1.9	0.1	-
U3 - Beta(1, 1)	10.6	10.1	5.9	1.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	5.2	1.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.8	4.7	1.9	0.7	0	-
U2 - Beta(0.7, 0.3)	2.8	4.7	1.6	0.8	0.1	-
U2 - Beta(1, 1)	2.6	4.8	1.9	0.7	0	-
U3 - Beta(0.3, 0.7)	3.4	5.2	1.7	0.1	0	-
U3 - Beta(0.5, 0.5)	2.9	4.7	1.8	0.7	0	-
U3 - Beta(0.7, 0.3)	2.9	4.8	1.6	0.8	0	-
U3 - Beta(1, 1)	2.9	4.8	1.8	0.7	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						970.6
U2 - Beta(0.5, 0.5)						990.3
U2 - Beta(0.7, 0.3)						999.5
U2 - Beta(1, 1)						993.5
U3 - Beta(0.3, 0.7)						964.5
U3 - Beta(0.5, 0.5)						989.9
U3 - Beta(0.7, 0.3)						994.4
U3 - Beta(1, 1)						988.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.5	37.8	39.9	1.5	0.1	7.2
U2 - Beta(0.5, 0.5)	16.5	34.1	39.1	3.7	0	6.6
U2 - Beta(0.7, 0.3)	18.2	34.8	35.9	4.5	0	6.6
U2 - Beta(1, 1)	16.7	34	38.8	4.4	0	6.1
U3 - Beta(0.3, 0.7)	19.6	39.7	32.1	1.5	0	7.1
U3 - Beta(0.5, 0.5)	19.9	34.4	35.8	3.1	0	6.8
U3 - Beta(0.7, 0.3)	20.3	34.7	34.5	4.2	0	6.3
U3 - Beta(1, 1)	19.7	34.1	36.2	3.2	0	6.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	11.9	8.3	0.7	0	-
U2 - Beta(0.5, 0.5)	7.6	9.8	9.3	2.3	0.2	-
U2 - Beta(0.7, 0.3)	7.8	9.7	8.3	3	0.4	-
U2 - Beta(1, 1)	7.6	9.8	9.4	2.4	0.2	-
U3 - Beta(0.3, 0.7)	9.2	12.3	7.1	0.5	0	-
U3 - Beta(0.5, 0.5)	8	10	9	2.1	0.1	-
U3 - Beta(0.7, 0.3)	8.1	9.8	8.2	2.9	0.2	-
U3 - Beta(1, 1)	8	9.9	9	2.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	5.3	2.8	0.3	0	-
U2 - Beta(0.5, 0.5)	1.8	4.6	2.8	0.9	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.6	2.4	1	0.1	-
U2 - Beta(1, 1)	1.7	4.5	2.8	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.6	5.5	2.4	0.2	0	-
U3 - Beta(0.5, 0.5)	2	4.7	2.7	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2	4.6	2.4	1	0.1	-
U3 - Beta(1, 1)	1.9	4.6	2.7	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021.2
U2 - Beta(0.5, 0.5)						1051.2
U2 - Beta(0.7, 0.3)						1054.8
U2 - Beta(1, 1)						1050.8
U3 - Beta(0.3, 0.7)						1013.2
U3 - Beta(0.5, 0.5)						1045.2
U3 - Beta(0.7, 0.3)						1052
U3 - Beta(1, 1)						1044.5

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.7	27.7	1.2	0	0	56.4
U2 - Beta(0.5, 0.5)	18.2	24.2	1.3	0	0	56.3
U2 - Beta(0.7, 0.3)	18.1	25.1	1.3	0	0	55.5
U2 - Beta(1, 1)	16.8	25.4	1.5	0	0	56.3
U3 - Beta(0.3, 0.7)	15	27.5	1	0	0	56.5
U3 - Beta(0.5, 0.5)	18.2	24.4	1.4	0	0	56
U3 - Beta(0.7, 0.3)	17.8	25.1	1.6	0	0	55.5
U3 - Beta(1, 1)	17.9	24.7	1.4	0	0	56
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	10.1	1.1	0.1	0	-
U2 - Beta(0.5, 0.5)	11.6	8.9	2.1	0.4	0	-
U2 - Beta(0.7, 0.3)	11.5	8.7	2.2	0.6	0	-
U2 - Beta(1, 1)	11.6	8.9	2.1	0.4	0	-
U3 - Beta(0.3, 0.7)	11.6	10.1	1	0.1	0	-
U3 - Beta(0.5, 0.5)	11.7	8.9	2	0.4	0	-
U3 - Beta(0.7, 0.3)	11.7	8.8	2.1	0.5	0	-
U3 - Beta(1, 1)	11.7	9	2	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	4.6	0.4	0	0	-
U2 - Beta(0.5, 0.5)	3.1	4.2	0.7	0.2	0	-
U2 - Beta(0.7, 0.3)	3.1	4.1	0.7	0.3	0	-
U2 - Beta(1, 1)	3.1	4.2	0.7	0.2	0	-
U3 - Beta(0.3, 0.7)	3.1	4.6	0.4	0	0	-
U3 - Beta(0.5, 0.5)	3.1	4.2	0.6	0.2	0	-
U3 - Beta(0.7, 0.3)	3.1	4.1	0.7	0.2	0	-
U3 - Beta(1, 1)	3.1	4.2	0.6	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						799
U2 - Beta(0.5, 0.5)						814.9
U2 - Beta(0.7, 0.3)						819.4
U2 - Beta(1, 1)						815
U3 - Beta(0.3, 0.7)						797.4
U3 - Beta(0.5, 0.5)						812.4
U3 - Beta(0.7, 0.3)						816.6
U3 - Beta(1, 1)						814.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	87.5	9.3	0.1	0	0	3.1
U2 - Beta(0.5, 0.5)	70.5	25.1	1	0	0	3.4
U2 - Beta(0.7, 0.3)	70.9	24.6	1.1	0	0	3.4
U2 - Beta(1, 1)	70.4	24.9	1.2	0	0	3.5
U3 - Beta(0.3, 0.7)	87.9	9.1	0	0	0	3
U3 - Beta(0.5, 0.5)	72.2	23.5	0.7	0	0	3.6
U3 - Beta(0.7, 0.3)	71.1	23.9	1.4	0	0	3.6
U3 - Beta(1, 1)	74.4	22	0.4	0	0	3.2
Mean administrations						
U2 - Beta(0.3, 0.7)	26.4	3	0.1	0	0	-
U2 - Beta(0.5, 0.5)	20.2	8.2	0.9	0.1	0	-
U2 - Beta(0.7, 0.3)	18.7	8.6	1.7	0.4	0	-
U2 - Beta(1, 1)	20.1	8.3	0.9	0.1	0	-
U3 - Beta(0.3, 0.7)	26.4	2.9	0	0	0	-
U3 - Beta(0.5, 0.5)	20.7	8	0.6	0	0	-
U3 - Beta(0.7, 0.3)	18.9	8.8	1.4	0.2	0	-
U3 - Beta(1, 1)	20.9	7.9	0.6	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7.5	1.2	0	0	0	-
U2 - Beta(0.5, 0.5)	5	3.1	0.3	0	0	-
U2 - Beta(0.7, 0.3)	4.6	3.1	0.5	0.1	0	-
U2 - Beta(1, 1)	5	3.2	0.3	0	0	-
U3 - Beta(0.3, 0.7)	7.5	1.2	0	0	0	-
U3 - Beta(0.5, 0.5)	5.3	3.1	0.2	0	0	-
U3 - Beta(0.7, 0.3)	4.6	3.2	0.5	0.1	0	-
U3 - Beta(1, 1)	5.3	3.1	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1140.2
U2 - Beta(0.5, 0.5)						1151
U2 - Beta(0.7, 0.3)						1159.8
U2 - Beta(1, 1)						1151.2
U3 - Beta(0.3, 0.7)						1140.6
U3 - Beta(0.5, 0.5)						1148.1
U3 - Beta(0.7, 0.3)						1152.8
U3 - Beta(1, 1)						1150.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.9	18	26.1	44.2	1.4	0.4
U2 - Beta(0.5, 0.5)	14.2	16.1	19.1	43.9	6.2	0.5
U2 - Beta(0.7, 0.3)	16.4	18.5	18.4	41.1	5.2	0.4
U2 - Beta(1, 1)	14.8	15.1	18.3	45.2	6.2	0.4
U3 - Beta(0.3, 0.7)	16.5	23.3	24.2	35.2	0.4	0.4
U3 - Beta(0.5, 0.5)	19.7	19.2	16.1	39.7	4.8	0.5
U3 - Beta(0.7, 0.3)	22.3	20.4	15.7	36.9	4.3	0.4
U3 - Beta(1, 1)	21.3	19.3	15.2	39.5	4.3	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6.2	8.2	8.2	7	0.4	-
U2 - Beta(0.5, 0.5)	4.7	6.1	6.9	9.9	2.4	-
U2 - Beta(0.7, 0.3)	4.9	5.9	6.8	9.1	3.2	-
U2 - Beta(1, 1)	4.7	6.1	6.9	9.9	2.4	-
U3 - Beta(0.3, 0.7)	7.2	8.9	7.7	5.9	0.2	-
U3 - Beta(0.5, 0.5)	4.9	6.4	6.9	9.7	2.1	-
U3 - Beta(0.7, 0.3)	5.2	6.3	6.8	8.9	2.6	-
U3 - Beta(1, 1)	5.1	6.6	6.8	9.5	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	4.1	3.2	2.4	0.2	-
U2 - Beta(0.5, 0.5)	0.7	3.4	2.3	3.4	0.8	-
U2 - Beta(0.7, 0.3)	0.8	3.5	2.1	3.1	1	-
U2 - Beta(1, 1)	0.7	3.4	2.3	3.3	0.8	-
U3 - Beta(0.3, 0.7)	1.9	4.4	3	2.1	0.1	-
U3 - Beta(0.5, 0.5)	0.8	3.6	2.3	3.3	0.7	-
U3 - Beta(0.7, 0.3)	1	3.6	2.2	3	0.9	-
U3 - Beta(1, 1)	0.9	3.7	2.2	3.2	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022.4
U2 - Beta(0.5, 0.5)						1065
U2 - Beta(0.7, 0.3)						1073.1
U2 - Beta(1, 1)						1065.3
U3 - Beta(0.3, 0.7)						1010.4
U3 - Beta(0.5, 0.5)						1060
U3 - Beta(0.7, 0.3)						1064.8
U3 - Beta(1, 1)						1058.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.9	77.5	9	0.5	0.1	0
U2 - Beta(0.5, 0.5)	1.4	54.3	32.1	11.7	0.5	0
U2 - Beta(0.7, 0.3)	3.9	36.7	27.3	25.2	6.9	0
U2 - Beta(1, 1)	1.3	53.5	33.4	11.3	0.5	0
U3 - Beta(0.3, 0.7)	15.4	78.7	5.5	0.4	0	0
U3 - Beta(0.5, 0.5)	2	59.9	27.9	9.8	0.4	0
U3 - Beta(0.7, 0.3)	3.1	43.8	28.6	22.3	2.2	0
U3 - Beta(1, 1)	1.8	59.6	28.2	10.2	0.2	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	18.3	1.7	0.1	0	-
U2 - Beta(0.5, 0.5)	4.1	16.3	7.3	2.2	0.2	-
U2 - Beta(0.7, 0.3)	3.7	8.3	8.2	7.4	2.4	-
U2 - Beta(1, 1)	4	16	7.5	2.3	0.2	-
U3 - Beta(0.3, 0.7)	10.2	18.6	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	4.2	17.1	6.6	2	0.1	-
U3 - Beta(0.7, 0.3)	3.9	11	7.8	5.9	1.3	-
U3 - Beta(1, 1)	4.2	17	6.7	2	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	6.6	0.6	0	0	-
U2 - Beta(0.5, 0.5)	0.4	6.1	2.3	0.7	0.1	-
U2 - Beta(0.7, 0.3)	0.3	3.5	2.3	2.1	0.8	-
U2 - Beta(1, 1)	0.3	6	2.4	0.7	0.1	-
U3 - Beta(0.3, 0.7)	3.2	6.7	0.4	0	0	-
U3 - Beta(0.5, 0.5)	0.4	6.3	2.1	0.6	0	-
U3 - Beta(0.7, 0.3)	0.3	4.3	2.3	1.8	0.4	-
U3 - Beta(1, 1)	0.4	6.3	2.1	0.7	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1092
U2 - Beta(0.5, 0.5)						1132.2
U2 - Beta(0.7, 0.3)						1159.4
U2 - Beta(1, 1)						1132.4
U3 - Beta(0.3, 0.7)						1090
U3 - Beta(0.5, 0.5)						1130.7
U3 - Beta(0.7, 0.3)						1149.4
U3 - Beta(1, 1)						1130
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	2.7	11.6	52.4	30.3	2.9	0.1
U2 - Beta(0.5, 0.5)	9.5	13.7	35	33.7	7.9	0.2
U2 - Beta(0.7, 0.3)	9.7	15.5	34.3	32.1	8.2	0.2
U2 - Beta(1, 1)	10.5	13.4	35.2	32.5	8.2	0.2
U3 - Beta(0.3, 0.7)	6.1	16	50.1	26.5	1.2	0.1
U3 - Beta(0.5, 0.5)	13.6	18	34.5	28.6	5.2	0.1
U3 - Beta(0.7, 0.3)	13.2	19.9	33.8	28.1	4.8	0.2
U3 - Beta(1, 1)	15.4	17.9	34.8	27.6	4.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	7.3	12.2	5.4	0.6	-
U2 - Beta(0.5, 0.5)	3.5	4.6	9.1	9.1	3.6	-
U2 - Beta(0.7, 0.3)	3.6	4.7	8.7	8.8	4.2	-
U2 - Beta(1, 1)	3.6	4.8	9.1	9	3.5	-
U3 - Beta(0.3, 0.7)	4.7	8	12.1	4.9	0.5	-
U3 - Beta(0.5, 0.5)	3.6	5	9.5	8.8	3	-
U3 - Beta(0.7, 0.3)	3.8	5.2	9	8.5	3.5	-
U3 - Beta(1, 1)	3.8	5.2	9.5	8.6	2.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.9	4.7	2	0.2	-
U2 - Beta(0.5, 0.5)	0.2	3	3	3.4	1.3	-
U2 - Beta(0.7, 0.3)	0.3	3	2.8	3.3	1.4	-
U2 - Beta(1, 1)	0.2	3	3	3.4	1.2	-
U3 - Beta(0.3, 0.7)	0.8	4.1	4.6	1.9	0.2	-
U3 - Beta(0.5, 0.5)	0.3	3.1	3.2	3.3	1	-
U3 - Beta(0.7, 0.3)	0.3	3.2	2.9	3.2	1.2	-
U3 - Beta(1, 1)	0.3	3.3	3.1	3.2	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1016
U2 - Beta(0.5, 0.5)						1050.3
U2 - Beta(0.7, 0.3)						1056.4
U2 - Beta(1, 1)						1049.9
U3 - Beta(0.3, 0.7)						1011.4
U3 - Beta(0.5, 0.5)						1046.3
U3 - Beta(0.7, 0.3)						1050.1
U3 - Beta(1, 1)						1046

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	5.1	24.1	59	7.2	0
U2 - Beta(0.5, 0.5)	8.8	6.7	15.9	53	15.5	0.1
U2 - Beta(0.7, 0.3)	10.1	9	16.1	48.7	16	0.1
U2 - Beta(1, 1)	9.5	7.1	15.4	52.3	15.6	0.1
U3 - Beta(0.3, 0.7)	7.6	5.3	24.6	58.5	3.9	0.1
U3 - Beta(0.5, 0.5)	12.7	9	14.6	50.9	12.7	0.1
U3 - Beta(0.7, 0.3)	13.5	10.4	13.9	47	15.1	0.1
U3 - Beta(1, 1)	13.1	9.3	12.9	52.4	12.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.8	9.3	10.9	1.2	-
U2 - Beta(0.5, 0.5)	3.3	3.9	6.6	11.6	4.5	-
U2 - Beta(0.7, 0.3)	3.3	3.8	6.4	10.5	5.9	-
U2 - Beta(1, 1)	3.4	3.9	6.6	11.6	4.5	-
U3 - Beta(0.3, 0.7)	3.9	5	9.6	10.8	0.8	-
U3 - Beta(0.5, 0.5)	3.4	4	6.7	11.8	4.1	-
U3 - Beta(0.7, 0.3)	3.6	4.1	6.5	10.6	5.2	-
U3 - Beta(1, 1)	3.6	4.2	6.6	11.7	3.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	3	3.5	3.8	0.4	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.9	4.1	1.3	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.9	3.8	1.6	-
U2 - Beta(1, 1)	0.2	2.8	1.9	4.1	1.3	-
U3 - Beta(0.3, 0.7)	0.4	3.1	3.6	3.8	0.3	-
U3 - Beta(0.5, 0.5)	0.2	2.8	2	4.2	1.2	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.9	3.8	1.5	-
U3 - Beta(1, 1)	0.3	2.9	1.9	4.1	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1038.8
U2 - Beta(0.5, 0.5)						1080.9
U2 - Beta(0.7, 0.3)						1087.8
U2 - Beta(1, 1)						1081.4
U3 - Beta(0.3, 0.7)						1033.7
U3 - Beta(0.5, 0.5)						1077.7
U3 - Beta(0.7, 0.3)						1080.1
U3 - Beta(1, 1)						1075.4
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10	17	56.7	10.5	0.6	5.2
U2 - Beta(0.5, 0.5)	14.4	19.4	47.6	12.5	1.6	4.5
U2 - Beta(0.7, 0.3)	14	19.9	47.4	12.2	1.6	4.9
U2 - Beta(1, 1)	14.1	18.1	47.7	13.4	1.8	4.9
U3 - Beta(0.3, 0.7)	12.8	20.2	51.4	10.4	0.2	5
U3 - Beta(0.5, 0.5)	17.7	20.1	43.6	12.7	1	4.9
U3 - Beta(0.7, 0.3)	17.3	21.7	42.5	13.2	1	4.3
U3 - Beta(1, 1)	19.2	21.3	40.5	13.8	0.6	4.6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.8	8.2	12.3	3.2	0.2	-
U2 - Beta(0.5, 0.5)	5.6	7.7	10.3	5	1.1	-
U2 - Beta(0.7, 0.3)	5.7	7.6	9.7	5.2	1.5	-
U2 - Beta(1, 1)	5.7	7.7	10.3	5	1	-
U3 - Beta(0.3, 0.7)	6.2	8.6	11.8	2.9	0.1	-
U3 - Beta(0.5, 0.5)	5.9	7.8	10.1	4.8	0.9	-
U3 - Beta(0.7, 0.3)	6.1	7.9	9.4	5	1.2	-
U3 - Beta(1, 1)	6.3	7.9	9.8	4.8	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	4	4.3	1.2	0.1	-
U2 - Beta(0.5, 0.5)	1.1	4.1	3.3	1.9	0.4	-
U2 - Beta(0.7, 0.3)	1.1	4.1	3.1	1.9	0.5	-
U2 - Beta(1, 1)	1.1	4.1	3.3	1.9	0.4	-
U3 - Beta(0.3, 0.7)	1.5	4.2	4.2	1.1	0	-
U3 - Beta(0.5, 0.5)	1.3	4.1	3.2	1.8	0.4	-
U3 - Beta(0.7, 0.3)	1.3	4.2	3	1.8	0.5	-
U3 - Beta(1, 1)	1.4	4.2	3.1	1.8	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1024.3
U2 - Beta(0.5, 0.5)						1037.3
U2 - Beta(0.7, 0.3)						1038.9
U2 - Beta(1, 1)						1036.4
U3 - Beta(0.3, 0.7)						1019.2
U3 - Beta(0.5, 0.5)						1034.6
U3 - Beta(0.7, 0.3)						1033.5
U3 - Beta(1, 1)						1031.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.6	75.6	10.5	0.2	0	0.1
U2 - Beta(0.5, 0.5)	5	57.6	32.8	4.3	0.2	0.1
U2 - Beta(0.7, 0.3)	7	49	32.2	11.2	0.5	0.1
U2 - Beta(1, 1)	5.6	56	33.6	4.5	0.2	0.1
U3 - Beta(0.3, 0.7)	16.5	75.6	7.4	0.4	0	0.1
U3 - Beta(0.5, 0.5)	6.5	62.9	26.8	3.7	0	0.1
U3 - Beta(0.7, 0.3)	7.2	54.1	29.5	8.9	0.2	0.1
U3 - Beta(1, 1)	7.4	62.7	26.8	3	0	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10	18	1.9	0.1	0	-
U2 - Beta(0.5, 0.5)	4.5	15.7	8	1.7	0.2	-
U2 - Beta(0.7, 0.3)	4.4	11.2	8.9	4.4	1	-
U2 - Beta(1, 1)	4.5	15.5	8.1	1.6	0.2	-
U3 - Beta(0.3, 0.7)	10.3	18.1	1.5	0.1	0	-
U3 - Beta(0.5, 0.5)	4.8	16.5	7.2	1.4	0.1	-
U3 - Beta(0.7, 0.3)	4.6	12.9	8.6	3.4	0.5	-
U3 - Beta(1, 1)	4.8	16.7	7	1.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	6.7	0.6	0	0	-
U2 - Beta(0.5, 0.5)	0.5	5.9	2.3	0.6	0.1	-
U2 - Beta(0.7, 0.3)	0.5	4.4	2.5	1.3	0.3	-
U2 - Beta(1, 1)	0.6	5.9	2.4	0.6	0.1	-
U3 - Beta(0.3, 0.7)	3.2	6.7	0.5	0	0	-
U3 - Beta(0.5, 0.5)	0.7	6.3	2.1	0.5	0	-
U3 - Beta(0.7, 0.3)	0.6	5	2.5	1.1	0.1	-
U3 - Beta(1, 1)	0.7	6.4	2	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1080.7
U2 - Beta(0.5, 0.5)						1134.6
U2 - Beta(0.7, 0.3)						1153.6
U2 - Beta(1, 1)						1134.4
U3 - Beta(0.3, 0.7)						1078.6
U3 - Beta(0.5, 0.5)						1129.4
U3 - Beta(0.7, 0.3)						1146.6
U3 - Beta(1, 1)						1128.4
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.3	36.6	11.1	1.2	0	35.8
U2 - Beta(0.5, 0.5)	17	36	10.1	2.1	0.1	34.7
U2 - Beta(0.7, 0.3)	17.4	37.2	10.1	1.6	0.1	33.6
U2 - Beta(1, 1)	16.1	37.8	9.8	1.9	0.2	34.2
U3 - Beta(0.3, 0.7)	17.4	36.3	8.8	1.1	0	36.4
U3 - Beta(0.5, 0.5)	19.5	37.4	7.3	1.5	0.2	34.1
U3 - Beta(0.7, 0.3)	18	38.1	8	1.4	0.2	34.3
U3 - Beta(1, 1)	19.6	36.8	7.3	1.8	0.2	34.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	11.2	3.9	0.9	0.1	-
U2 - Beta(0.5, 0.5)	9.6	10	4.3	1.9	0.3	-
U2 - Beta(0.7, 0.3)	9.6	9.9	4.3	2.1	0.4	-
U2 - Beta(1, 1)	9.6	10.1	4.3	1.9	0.3	-
U3 - Beta(0.3, 0.7)	10.2	11.3	3.5	0.8	0	-
U3 - Beta(0.5, 0.5)	9.9	10.1	4.1	1.8	0.3	-
U3 - Beta(0.7, 0.3)	9.9	9.9	4.1	2	0.3	-
U3 - Beta(1, 1)	10	10.1	4	1.8	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	5.3	1.4	0.4	0	-
U2 - Beta(0.5, 0.5)	2.4	5	1.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.4	5	1.3	0.9	0.1	-
U2 - Beta(1, 1)	2.4	5	1.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.7	5.3	1.3	0.4	0	-
U3 - Beta(0.5, 0.5)	2.5	5	1.3	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.5	5	1.3	0.8	0.1	-
U3 - Beta(1, 1)	2.6	5	1.3	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						883.3
U2 - Beta(0.5, 0.5)						903.5
U2 - Beta(0.7, 0.3)						908.9
U2 - Beta(1, 1)						905.2
U3 - Beta(0.3, 0.7)						879.3
U3 - Beta(0.5, 0.5)						901.2
U3 - Beta(0.7, 0.3)						904.2
U3 - Beta(1, 1)						903.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.9 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.6	45.8	9.2	0.1	0.1	27.2
U2 - Beta(0.5, 0.5)	17.6	47.3	7.9	0.3	0	26.9
U2 - Beta(0.7, 0.3)	17.8	46.6	8.5	0.4	0	26.7
U2 - Beta(1, 1)	18.8	45.6	8.4	0.3	0	26.9
U3 - Beta(0.3, 0.7)	19.7	45.5	7.6	0.2	0	27
U3 - Beta(0.5, 0.5)	20.6	44.5	7.1	0.4	0	27.4
U3 - Beta(0.7, 0.3)	19.6	45	8	0.4	0	27
U3 - Beta(1, 1)	19.9	44.9	8.2	0.4	0	26.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	12.8	4	0.8	0.1	-
U2 - Beta(0.5, 0.5)	9.4	11.6	4.4	1.7	0.2	-
U2 - Beta(0.7, 0.3)	9.3	11.3	4.5	2	0.3	-
U2 - Beta(1, 1)	9.2	11.8	4.4	1.7	0.2	-
U3 - Beta(0.3, 0.7)	9.9	13.1	3.5	0.7	0	-
U3 - Beta(0.5, 0.5)	9.7	11.6	4.2	1.7	0.2	-
U3 - Beta(0.7, 0.3)	9.6	11.5	4.2	1.8	0.3	-
U3 - Beta(1, 1)	9.7	11.7	4.2	1.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	5.9	1.5	0.4	0	-
U2 - Beta(0.5, 0.5)	2.3	5.7	1.4	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.2	5.6	1.4	0.8	0.1	-
U2 - Beta(1, 1)	2.2	5.7	1.4	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.7	6	1.3	0.3	0	-
U3 - Beta(0.5, 0.5)	2.4	5.7	1.3	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.4	5.7	1.3	0.8	0.1	-
U3 - Beta(1, 1)	2.4	5.7	1.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						925.2
U2 - Beta(0.5, 0.5)						939
U2 - Beta(0.7, 0.3)						940.4
U2 - Beta(1, 1)						940.8
U3 - Beta(0.3, 0.7)						924.4
U3 - Beta(0.5, 0.5)						937.9
U3 - Beta(0.7, 0.3)						938.3
U3 - Beta(1, 1)						938.3